Lecture 2: Random Variables

ECE 315

This chapter introduces the concept of random variables and presents several examples of commonly encountered discrete and continuous types. Discrete random variables are described using probability mass functions (PMFs), which effectively capture the behavior of variables with countable outcomes. However, PMFs are insufficient for variables that take on values from a continuous range. To address this limitation, the chapter introduces the cumulative distribution function (CDF) as a unified approach for describing both discrete and continuous random variables. For continuous cases specifically, the probability density function (PDF) is introduced as a key analytical tool.

Special attention is given to the widely used Gaussian (normal) distribution, highlighting its fundamental properties and practical relevance. The chapter also covers the concepts of conditional distributions and conditional density functions, which are crucial for analyzing relationships between random variables. To reinforce these ideas, a variety of practical examples are included to demonstrate their application in real-world scenarios.

The content of this lecture note is based mainly on Chapter 3 from Probability and Random Processes With Applications to Signal Processing and Communications, Edition 2, by Scott L.Millerand Chapter 3 from Probability and statistics for Engineers and Scientists, Edition 9, by Ronald Walpole, et al..

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1 Concept of a Random Variable

Definition 1 (Random Variable (RV))**.** A random variable is a real valued function of the elements of a sample space, S. Given an experiment, E, with sample space, S, the random variable X maps each possible outcome, ξ, to a real number as specified by some rule. If the mapping is such that the random variable takes on a finite or countably infinite number of values, then we refer to as a **discrete random variable**; whereas, if the range of is an uncountably infinite number of points, we refer to as a **continuous random variable**.

Example 1.1. The experiment of rolling a die and tossing a coin. Let X be a random variable that maps the **head** outcome to the positive number which correspond to the dots on the die, and **tail** outcome to the negative number that is double the number which appears on t he die. This mapping is illustrated in Figure 1

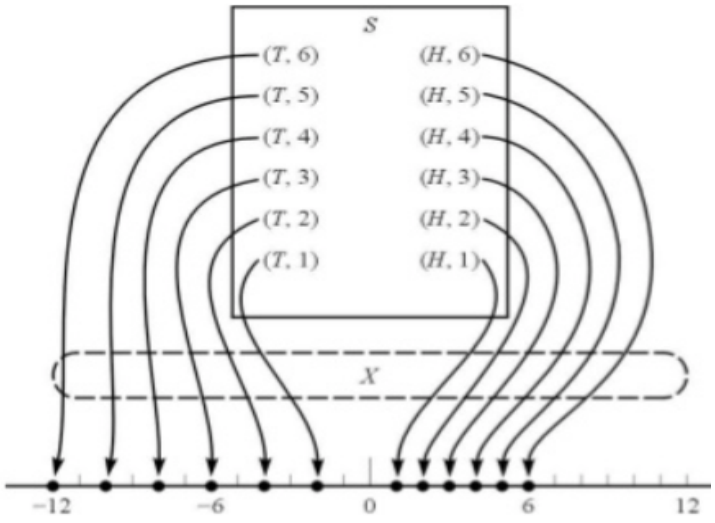


Figure 1: The experiment of rolling a die and tossing a coin.

Example 1.2. Consider the experiment of flipping a coin 3 times and counting the number of heads. Let the random variable X represents the number of heads. The sample space of this experiment is composed of 8 outcomes $S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$

- Equivalency between experiment and random variables
  Table 1 summarizes the analogy between random experiment and random variables.

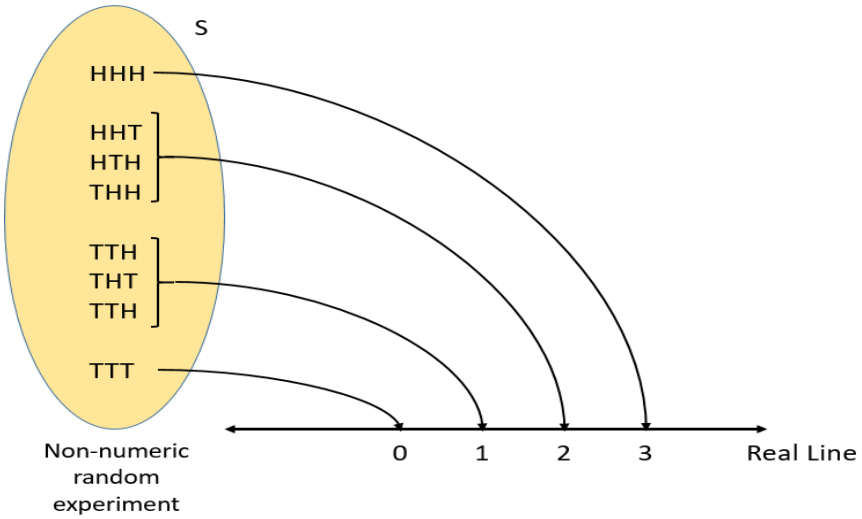


Figure 2: Mapping the experiment of flipping a coin 3 times to a random variable

Table 1: Equivalency between experiment and random variables

Unstructured Experiment	**Random Variable**
Sample space	Range of X
Outcome of E	One possible value x for X
Event	Subset of range of X
Event A	$x \in subsetofrangeof$X($e.g.$,x=3or2\leq x\leq 4$
Pr(A)	$Pr(X = 3),\ Pr(2 \leq X \leq 4)$

- Examples:

Example 1.3. Consider the simple condition in which components are arriving from the production line and they are stipulated to be defective or not defective.
Solution: Define the random variable X by

$$X = \begin{cases} 1 & \text{if the component is defective} \\ 0 & \text{if the component is non-defective} \end{cases} \tag{1}$$

Note: The random variable for which 0 and 1 are chosen to describe the two possible values is called a **Bernoulli random variable**.

Example 1.4. Statisticians use **sampling plans** to either accept or reject batches or lots of material. Suppose one of these sampling plans involves sampling independently 10 items from a lot of 100 items in which 12 are defective.
Solution: Let X be the random variable defined as the number of items found defective in the sample of 10. In this case, the random variable takes on the values $0, 1, 2, \ldots, 9, 10$

Example 1.5. Suppose a sampling plan involves sampling items from a process until a defective is observed. The evaluation of the process will depend on how many consecutive items are observed.
Solution: In that regard, let X be a random variable defined by the number of items observed before a defective is found. With N a nondefective and D a defective, mapping between the possible outcomes of the experiment and their corresponding value of the random variable X is explained in Table 2

Table 2: Example 1.5

Sample space	**x**
D	1
ND	2
NND	3
NNND	4
⋮	⋮

- Conditions for a function ot be a random variable
  - The function is not multivalued in the sense that each element of S is mapped to a single value in the domain of the random variable.
  - The set $\{X \leq x\}$ for any value of x is an event that is corresponding to the points s in the sample space, for which the random variable $X(s)$ does not exceeds the value of x.
- We use the convention that upper case variables represent random variables while lower case variables represent fixed values that the random variable can assume.
- Since $X = g(\xi)$ is a random variable whose numerical value depends on the outcome of an experiment, we cannot describe the random variable by stating its value; rather, we must give it a probabilistic description by stating the probabilities that the variable takes on a specific value or values (e.g.,$\Pr(X = 3)$ or $\Pr(X > 8)$). There are three common functions to describe the probabilistic behavior of a RV:
  - **Probability Mass Function (PMF):** for discrete RVs.
  - **Probability Density Function (PDF):** for continuous RVs.
  - **Cumulative Distribution Function (CDF):** for both discrete and continuous RVs.

2 Discrete Random Variables

- Although the definition of a discrete RV allows it to take any finite or countable set of values, the values are in practice almost integers.
- There are two main function used to describe the probabilistic behavior of discrete RVs:
  - Probability mass function.
  - Cumulative distribution function.

2.1 Probability Mass Function (PMF)

Definition 2 (Probability Mass Function (PMF))**.** The probability mass function (PMF), $P_X(x)$, of a random variable, X, is a function that assigns a probability to each possible value of the random variable, X . The probability that the random variable X takes on the specific value x_i is the value of the probability mass function for x_i. That is, $P_X(x_i) = \Pr(X = x_i)$.

Properties of the PMF:

1. $P_X(x_i) \geq 0$.
2. $\sum_{i=1} P_X(x_i) = 1$
3. $P_X(x_i) = \Pr(X = x_i)$
4. $\Pr(x_m \leq X \leq x_n) = \sum_{i=m}^{n} P_X(x_i)$

2.2 Cumulative Distribution Function (CDF)

Definition 3. The cumulative distribution function $F(x)$ of a discrete random variable X with probability mas function $P_X(x)$ is

$$F_X(x) = \Pr(X \leq x) = \sum_{i=1}^{k} P_X(x_i), for \quad x_k \leq x < x_{k+1} \tag{2}$$

The constraint in this equation can be incorporated using unit step functions, in which case the CDF of a discrete random variable can be written as:

$$F_X(x) = \sum_{i=1}^{k} P_X(x_i)\, u(x - x_i) \tag{3}$$

Properties of CDF

- $F_X(-\infty) = 0$.
- $F_X(\infty) = 1$.
- $0 \leq F_X(x) \leq 1$.
- The cumulative distribution function is a monotone non-decreasing function. That is, if $x_1 < x_2$, then $F_X(x_1) \leq F_X(x_2)$.

- $P(x_1 < X \leq x_2) = F_X(x_2) - F_X(x_1)$
- $F_X(x)$ is continuous from right. That is $F_X(x) = F_X(x^+)$.
- CDFs are defined not only for the values assumed by the given random variable but for all real numbers.

2.3 Plot of the PMF and CDF of a discrete RVs

2.3.1 PMF

According to the definition of the PMF, the PMF of a discrete RV is a train of impulses where these pulses occur at the values of the random variable (i.e., $x_i, \quad i = 1, 2, 3, \ldots, \ N$) and the value/amplitude of the pulse at x_i equals to the probability of the event $\{X = x_i\}$. Hence, we may write $f_X(x)$ as

$$f_X(x) = \sum_{i=1}^{N} P(x_i)\, \delta(x - x_i) \tag{4}$$

where $\delta(\cdot)$ is the delta function defined by

$$\delta(x - x_i) = \begin{cases} 1 & x = x_i \\ 0 & x \neq x_i \end{cases}$$

2.3.2 CDF

According to the definition and properties of CDF, the CDF of a discrete RV is a **staricase (stepwise)** function, where a step occurs at the values of the random variable (i.e., $x_i, \quad i = 1, 2, 3, \ldots, \ N$). N may be infinite for some random variables. The value of the CDF at each step (i.e., $F_X(x_i)$) equals to the probability of the event $\{X \leq x_i\}$, which is equal to the sum of the probability of all elementary events corresponding to $\{X \leq x_i\}$. Hence, we may write $F_X(x)$ as

$$F_X(x_n) = \sum_{i=1}^{n} P\{X = x_i\}\, u(x - x_i) \tag{5}$$

where $u(\cdot)$ is the unit-step function defined by

$$u(x) = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0 \end{cases}$$

2.4 Examples

Example 2.1. The Random Variable is X equals to the sum of the scores on the two dice. Find:

A) The possible values of the X.

B) The PMF of X.

C) The CDF of X.

D) The probability that the sum is less than or equal to 4.

E) The probability that the sum is greater than 7.

F) The probability that the sum is between 5 and 9.

G) Plot the PMF and CDF of X.

Solution:

A) The possible values of the X. There are $6 \times 6 = 36$ possible outcomes, and the possible values of the X (which is the sum of the scores on the two dice) are $\{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

B) To find the PMF, we need to find the probability of each possible value of X. To do this, we need to count how often each value occurs:

- 2 occurs just once, so $\Pr(X = 2) = 1/36$
- 3 occurs twice, so $\Pr(X = 3) = 2/36 = 1/18$
- 4 occurs three times, so $\Pr(X = 4) = 3/36 = 1/12$
- 5 occurs four times, so $\Pr(X = 5) = 4/36 = 1/9$
- 6 occurs five times, so $\Pr(X = 6) = 5/36$
- 7 occurs six times, so $\Pr(X = 7) = 6/36 = 1/6$
- 8 occurs five times, so $\Pr(X = 8) = 5/36$
- 9 occurs four times, so $\Pr(X = 9) = 4/36 = 1/9$
- 10 occurs three times, so $\Pr(X = 10) = 3/36 = 1/12$
- 11 occurs twice, so $\Pr(X = 11) = 2/36 = 1/18$
- 12 occurs just once, so $\Pr(X = 12) = 1/36$

The PMF is summarized in Table 3.

Table 3: Example 2.1

x	2	3	4	5	6	7	8	9	10	11	12
$P_X(x)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

C) Based on the definition of the CDF,

- $F_X(x) = 0 \; \forall \; x < 2$.

- $F_X(x) = \frac{1}{36}$; $2 \leq x < 3$.
- $F_X(x) = \frac{1}{36} + \frac{2}{36} = \frac{3}{36}$; $3 \leq x < 4$.
- $F_X(x) = \frac{1}{36} + \frac{2}{36} + \frac{3}{36} = \frac{6}{36}$; $4 \leq x < 5$.
- $F_X(x) = \frac{1}{36} + \frac{2}{36} + \frac{3}{36} + \frac{4}{36} = \frac{10}{36}$; $5 \leq x < 6$.
- $F_X(x) = \frac{1}{36} + \frac{2}{36} + \frac{3}{36} + \frac{4}{36} + \frac{5}{36} = \frac{15}{36}$; $6 \leq x < 7$.
- $F_X(x) = \frac{1}{36} + \frac{2}{36} + \frac{3}{36} + \frac{4}{36} + \frac{5}{36} + \frac{6}{36} = \frac{21}{36}$; $7 \leq x < 8$.
- $F_X(x) = \frac{1}{36} + \frac{2}{36} + \frac{3}{36} + \frac{4}{36} + \frac{5}{36} + \frac{6}{36} + \frac{5}{36} = \frac{26}{36}$; $8 \leq x < 9$.
- $F_X(x) = \frac{1}{36} + \frac{2}{36} + \frac{3}{36} + \frac{4}{36} + \frac{5}{36} + \frac{6}{36} + \frac{5}{36} + \frac{4}{36} = \frac{30}{36}$; $9 \leq x < 10$.
- $F_X(x) = \frac{1}{36} + \frac{2}{36} + \frac{3}{36} + \frac{4}{36} + \frac{5}{36} + \frac{6}{36} + \frac{5}{36} + \frac{4}{36} + \frac{3}{36} = \frac{33}{36}$; $10 \leq x < 11$.
- $F_X(x) = \frac{1}{36} + \frac{2}{36} + \frac{3}{36} + \frac{4}{36} + \frac{5}{36} + \frac{6}{36} + \frac{5}{36} + \frac{4}{36} + \frac{3}{36} + \frac{2}{36} = \frac{35}{36}$; $11 \leq x < 12$.
- $F_X(x) = \frac{1}{36} + \frac{2}{36} + \frac{3}{36} + \frac{4}{36} + \frac{5}{36} + \frac{6}{36} + \frac{5}{36} + \frac{4}{36} + \frac{3}{36} + \frac{2}{36} + \frac{1}{36} = \frac{36}{36} = 1$; $x \geq 12$.

D) Using the CDF, the probability that the sum is less than or equal 4 is given by

$$\Pr(X \leq 4) = F_X(X = 4) = \frac{6}{36}. \tag{6}$$

Alternatively, we can use the PMF to find the probability as follows:

$$\Pr(X \leq 4) = \Pr(X = 2) + \Pr(X = 3) + \Pr(X = 4) = \frac{1}{36} + \frac{2}{36} + \frac{3}{36} = \frac{6}{36} \tag{7}$$

E) Using the CDF, the probability that the sum is greater than 7 is given by

$$\Pr(X > 7) = 1 - \Pr(X \leq 7) = 1 - F_X(7) = 1 - \frac{21}{36} = \frac{15}{36} \tag{8}$$

Alternatively, we can use the PMF to find the probability as follows:

$$\begin{aligned} \Pr(X > 7) &= \Pr(X = 8) + \Pr(X = 9) + \Pr(X = 10) + \Pr(X = 11) + \Pr(X = 12) \\ &= \frac{5}{36} + \frac{4}{36} + \frac{3}{36} + \frac{2}{36} + \frac{1}{36} = \frac{15}{36} \end{aligned} \tag{9}$$

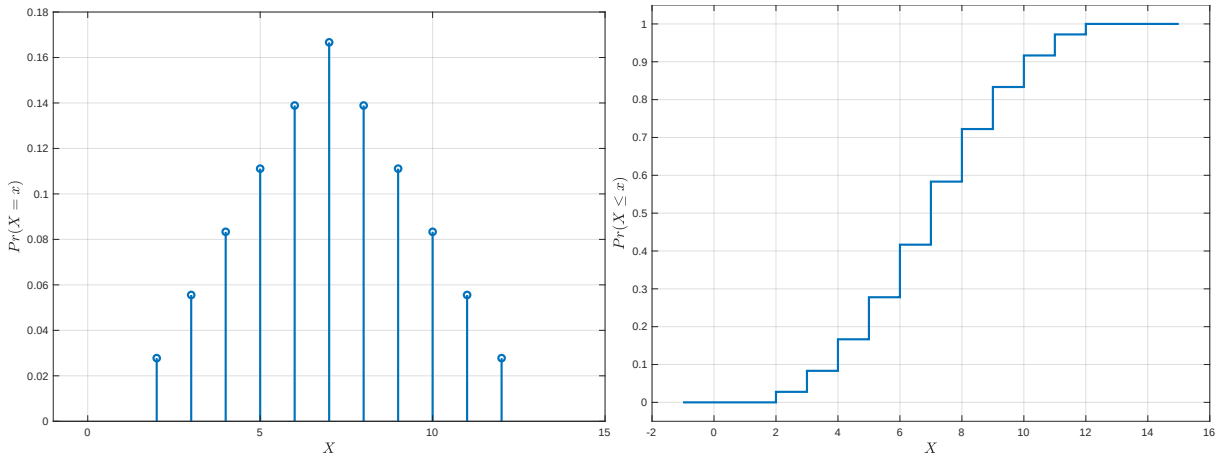
F) Using the CDF, the probability that the sum is between 5 and 9 is

$$\Pr(5 \leq X \leq 9) = \Pr(4 < X \leq 9) = F_X(9) - F_X(4) = \frac{30}{36} - \frac{6}{36} = \frac{24}{36} \tag{10}$$

Alternatively, we can use the PMF to find the probability as follows:

$$\begin{aligned} \Pr(5 \leq X \leq 9) &= \Pr(X = 5) + \Pr(X = 6) + \Pr(X = 7) + \Pr(X = 8) + \Pr(X = 9) \\ &= \frac{4}{36} + \frac{5}{36} + \frac{6}{36} + \frac{5}{36} + \frac{4}{36} = \frac{24}{36} \end{aligned} \tag{11}$$

G) The plot of the PMF and CDF is shown in Figure 3.



(a) PMF (b) CDF

Figure 3: PMF and CDF of the the random variable X in example 2.1

Example 2.2. A shipment of 20 similar laptop computers to a retail outlet contains 3 that are defective. If a school makes a random purchase of 2 of these computers, find the probability distribution for the number of defectives. Plot the corresponding PMF and CDF.
Solution: Let X be a random variable whose values x are the possible numbers of defective computers purchased by the school. Then x can only take the numbers $0, 1,$ and 2. Now

$$f(0) = \Pr(X = 0) = \frac{\binom{3}{0}\binom{17}{2}}{\binom{20}{2}} = \frac{136}{190}$$

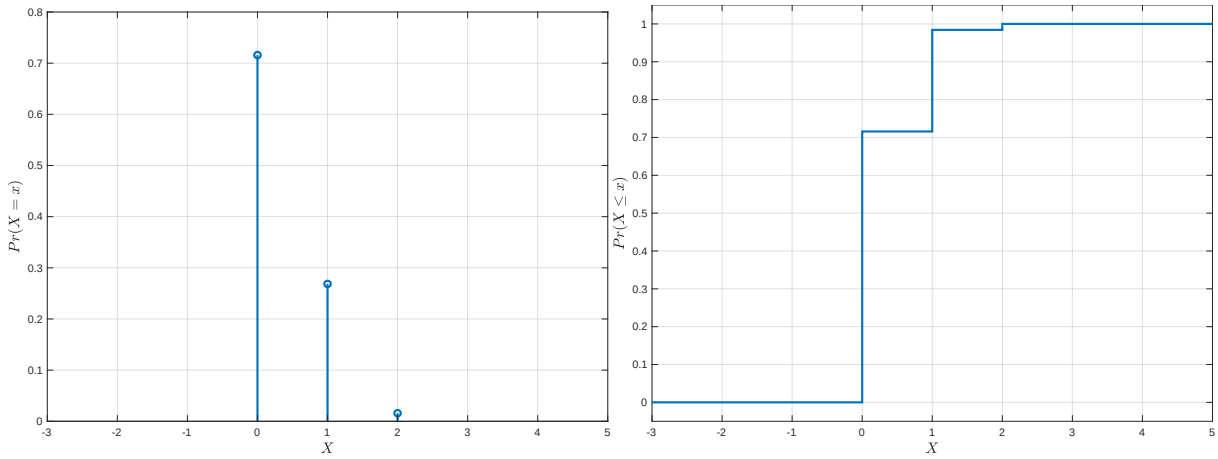
$$f(1) = \Pr(X = 1) = \frac{\binom{3}{1}\binom{17}{1}}{\binom{20}{2}} = \frac{51}{190}$$

$$f(2) = \Pr(X = 2) = \frac{\binom{3}{2}\binom{17}{0}}{\binom{20}{2}} = \frac{3}{190}$$

Thus, the probability distribution of X is

Table 4: Example 2.2

x	0	1	2
$P_X(x)$	$\frac{136}{190}$	$\frac{51}{190}$	$\frac{3}{190}$



(a) PMF (b) CDF

Figure 4: PMF and CDF of the the random variable X in example 2.2

3 Common Discrete Random Variables

3.1 Bernoulli Random Variable

- Description:
  Bernoulli random variable is the simplest possible random variable and can be defined formally as:

Definition 4. Bernoulli random variable is a discrete random variable that takes only two possible values:

$$X = \begin{cases} 1 & \text{with probability } p \\ 0 & \text{with probability } 1-p \end{cases}$$

where $0 \leq p \leq 1$.
Here, p represents the probability of **success**, and $1-p$ represents the probability of **failure**.

- **Probability Mass Function (PMF)**

$$P_X(x) = \begin{cases} p & \text{if } x = 1 \\ 1-p & \text{if } x = 0 \end{cases}$$

- **Practical Applications:**
  - **Digital Communication:** Each transmitted bit is either a 0 or 1 (success or failure in bit detection).
  - **Radar Systems:** A target can be wither presence (1) or absent (0.
  - **Quality Control:** An item is either defective (1) or not defective (0).

- **Medical Testing:** A patient either tests positive (1) or negative (0) for a condition.
- **Website A/B Testing:** A user either clicks on a link (1) or doesn't (0).
- **Survey Response:** A respondent answers "Yes" (1) or "No" (0) to a yes/no question.
- **Loan Default Prediction:** A borrower either defaults (1) or repays (0).
- **Online Purchases:** A customer either buys a product (1) or not (0).
- **Email Open Rate:** An email is either opened (1) or ignored (0).
- **Authentication Systems:** A login attempt is either accepted (1) or rejected (0).

3.2 Binomial Random Variable

- **Description:**
The **binomial distribution** models the number of successes in a fixed number of independent Bernoulli trials, where each trial has only two possible outcomes: success or failure. This distribution arises naturally when we repeat the same experiment multiple times under identical conditions.

Definition 5. Let X be a random variable representing the number of successes in n independent Bernoulli trials, each with probability of success p. Then X follows a binomial distribution with parameters n and p, denoted by:

$$X \sim \text{Binomial}(n, p)$$

- **Probability Mass Function (PMF)** The probability that the binomial random variable X takes a specific value k (i.e., exactly k successes) is given by the PMF:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad \text{for } k = 0, 1, 2, \ldots, n$$

Where:

- $\binom{n}{k} = \dfrac{n!}{k!(n-k)!}$ is the binomial coefficient
- p is the probability of success in one trial
- $1 - p$ is the probability of failure

- **Derivation of the Binomial PMF**
We perform n independent Bernoulli trials, each with:

- Probability of success: p
- Probability of failure: $1 - p$

Let X be the number of successes in the n trials. We want to compute $P(X = k)$, the probability of getting exactly k successes.

Probability of One Specific Sequence

Consider a specific sequence with k successes and $n-k$ failures. Since the trials are independent, the probability of this exact sequence is:

$$p^k(1-p)^{n-k}$$

Number of Such Sequences

The number of different sequences (arrangements) that contain exactly k successes out of n trials is given by the binomial coefficient:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Total Probability

To find the total probability of getting exactly k successes in n trials, we multiply the probability of one such sequence by the number of such sequences:

$$P_X(x) = \Pr(X = k) = \binom{n}{k} p^k(1-p)^{n-k}$$

- **Practical Applications:**
  - **Quality Control:** Counting the number of defective items in a batch of n products.
  - **Email Marketing:** Number of recipients who open an email in a campaign sent to n people.
  - **Clinical Trials:** Number of patients who respond to a new treatment among a group of n treated patients.
  - **Education:** Number of students passing a test assuming each has the same probability of passing.
  - **Networks:** Number of successful packet transmissions out of n attempts over a noisy channel.
  - **Voting:** Number of voters who support a candidate in a randomly sampled group of n voters.
  - **Cybersecurity:** Number of login attempts that are successful out of n attempts.

Example 3.1 (Email Marketing Campaign)**.** A company sends a promotional email to $n = 100$ customers. From previous data, it is known that each customer opens the email with a probability $p = 0.2$.

(a) What is the probability that exactly $k = 25$ customers open the email?

(b) What is the probability that 10 or fewer customers open the email?

(c) What is the probability that 15 or more customers open the email?

Solution:
Part a: What is the probability that is exactly "25" customers open the email? We use the binomial probability formula:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Substitute the given values:

$$P(X = 25) = \binom{100}{25} (0.2)^{25} (0.8)^{75}$$

Part (b): What is the probability that 10 or fewer customers open the email?

We want to find:

$$P(X \leq 10) = \sum_{k=0}^{10} \binom{100}{k} (0.2)^k (0.8)^{100-k}$$

This is the value of the cumulative distribution function (CDF) of the binomial distribution at $k = 10$. Using software (e.g., MATLAB or Python), we get:

$$P(X \leq 10) \approx 0.0057$$

Part (c): What is the probability that 15 or more customers open the email?

We want to find:

$$P(X \geq 15) = 1 - P(X \leq 14)$$

Using the binomial CDF:

$$P(X \leq 14) \approx 0.0804 \quad \Rightarrow \quad P(X \geq 15) \approx 1 - 0.0804 = 0.9196$$

These expression is computationally intensive to evaluate by hand, so it is best computed using statistical software. For example, using Matlab:

```
% Parameters
n = 100;        % number of trials
p = 0.2;        % probability of success

%% Part a: What is the probability that is exactly "k" customers open
    the email?
k = 25;         % number of successes

% Binomial probability mass function
```

```
prob_k = binopdf(k, n, p);

% Display the result
fprintf('The probability that exactly %i customers open the email out of %i customers is %.4f\n', k,n,prob_k);
%% Part b: What is the probability that "k" or fewer customers open the email?
k = 10;        % number of successes

% Binomial cumulative distribution function
prob_cdf = binocdf(k, n, p);

% Display the result
fprintf('The probability that %i or fewer customers open the email  out of %i customers is %.4f\n',k,n, prob_cdf);
%% Part c: What is the probability that "k" or more customers open the email?
k = 15;        % number of successes
% Binomial complementary cumulative distribution function
prob_ccdf = 1 - binocdf(k-1, n, p);

% Display the result
fprintf('The probability that more than or equal to %i open the email out of %i customers is %.4f\n',k,n, prob_ccdf);
```

3.3 Poisson Random Variable

- **Description:**
  The **Poisson random variable** models the number of times an event occurs in a fixed interval of time or space, given that the events happen independently and at a constant average rate. It is commonly used to model rare or random events.

Definition 6. A random variable X is said to follow a Poisson distribution with rate parameter $\lambda > 0$, if it represents the number of events in a fixed interval of time, distance, area, or volume, where:

- Events occur independently.
- The average rate λ (events per unit interval) is constant.
- Two events cannot occur at exactly the same instant.

Notation:

$$X \sim \text{Poisson}(\lambda)$$

- **Probability Mass Function (PMF)**

  The PMF of a Poisson random variable X is:

  $$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \quad \text{for } k = 0, 1, 2, \ldots$$

  Where:

  - λ: expected number of occurrences in the interval

    $$\lambda = \rho \times T$$

    where ρ represents the *average occurrence rate* and T represents the given interval
  - k: actual number of occurrences
  - $e \approx 2.71828$: Euler's number

- **Practical Applications:**

  - **Call Centers:** Number of incoming calls per minute at a customer service desk.
  - **Network Traffic:** Number of data packets arriving at a router per second.
  - **Natural Events:** Number of earthquakes in a region over a year.
  - **Healthcare:** Number of patients arriving at an emergency room per hour.
  - **Astronomy:** Number of photons hitting a detector in a given time.
  - **Web Services:** Number of hits to a website per minute.
  - **Manufacturing:** Number of defects per unit length in a roll of material.
  - **Insurance:** Number of claims filed per day.
  - **Transportation:** Number of cars passing through a toll booth per hour.
  - **Biology:** Number of mutations in a strand of DNA over a specific length.

- **Relationship with Binomial Distribution**

  The Poisson distribution arises frequently in practice because it serves as a good approximation to the binomial distribution when the number of trials n is large and the probability of success p is small.

  The binomial distribution Binomial(n, p) has two parameters and its PMF involves binomial coefficients, which become computationally intensive for large n. The Poisson distribution, on the other hand, is simpler: it depends on a single parameter λ, and does not involve binomial coefficients.

  Approximation Rule: Let $X_n \sim \text{Binomial}(n, p)$, and suppose that $n \to \infty$ and $p \to 0$ such that:

  $$\lim_{n\to\infty} np = \lambda$$

  Then the binomial distribution converges to a Poisson distribution:

  $$\text{Binomial}(n, p) \approx \text{Poisson}(\lambda)$$

This approximation is particularly useful in modeling rare events (small p) that occur over a large number of trials (large n), such as system failures, accidents, or mutations.

Example 3.2. The number of flaws in a fibre optic cable follows a Poisson distribution. The average number of flaws in 50 m of cable is 1.2.

a) What is the probability of exactly three flaws in 150m of cable?

b) What is the probability of at least two flaws in 100m of cable?

c) What is the probability of exactly one flaw in the first 50m of cable and exactly one flaw in the second 50m of cable?

Solution: From the given information, the *success rate* $\rho = 1.2/50 = 0.024$.

a) Mean number of flaws in $T = 150m$ of cable is $\lambda = \rho \times\ T = 3.6$. So the probability of exactly three flaws in 150m of cable is

$$\Pr(k) = e^{-\lambda}\frac{\lambda^k}{k!} = e^{-3.6}\frac{3.6^3}{3!} = 0.212. \tag{12}$$

b) Mean number of flaws in $T = 100m$ of cable is $\lambda = \rho \times T = 2.4$. So the probability of at least two flaws in 100 m of cable is

$$\Pr(X \geq 2) = 1 - \Pr(X \leq 1) = 1 - (\Pr(X = 0) + \Pr(X = 1)) \tag{13}$$

$$= 1 - (e^{-2.4}\frac{2.4^0}{0!} + e^{-2.4}\frac{2.4^1}{1!}) = 1 - (0.091 + 0.218) = 0.691 \tag{14}$$

c) Mean number of flaws in $50m$ of cable is $\lambda = 1.2$. So the probability of exactly one flaw in the first 50m is

$$\Pr(k) = e^{-1.2}\frac{1.2^1}{1!} = 0.361. \tag{15}$$

As X follows a Poisson distribution, the occurrence of flaws in the first and second 50m of cable are independent. Thus the probability of exactly one flaw in the first 50m and exactly one flaw in the second 50m is $(0.361)(0.361) = 0.13$.

Example 3.3. The number of visitors to a webserver per minute follows a Poisson distribution. If the average number of visitors per minute is 4, what is the probability that:

a) There are two or fewer visitors in one minute?

b) There are exactly two visitors in 30 seconds?

Solution:

a) We need the average number of visitors in a minute. In this case the parameter $\lambda = 4$. We wish to calculate:

$$\begin{aligned} \Pr(X \leq 2) &= P(X = 0) + P(X = 1) + P(X = 2) \\ &= e^{-4} + 4e^{-4} + 8e^{-4} = 0.238. \end{aligned} \tag{16}$$

b) If the average number of visitors in 1 minute is 4, the average in 30 seconds is 2. So for this example, our parameter $b = 2$. So

$$\Pr(X = 2) = e^{-2}\frac{2^2}{2!} = 0.271. \tag{17}$$

Note: The previous example is a standard example of a queueing process. These are very important in many applications in communications engineering and networking. Other examples of this type include the number of calls to an exchange, the arrival of customers at a service desk or the arrival of cars at a toll booth.

3.4 Geometric Random Variable

- **Description:**
  The **Geometric distribution** models the number of independent Bernoulli trials needed to obtain the **first success**. Each trial has two outcomes:
  - Success with probability p
  - Failure with probability $1 - p$

  The trials are repeated until the first success is observed.

  Example: Flipping a coin until the first heads appears.

  The geometric distribution is often used in ***waiting time problems***, answering the question: "How many trials do we need to wait until the first success?"

Definition 7. Let the random variable X represent the number of trials needed to obtain the first success. We say:

$$X \sim \text{Geometric}(p)$$

where:

- $p \in (0, 1]$ is the probability of success in each trial
- $X \in \{1, 2, 3, \ldots\}$

- **Probability Mass Function (PMF) and Cumulative Distribution Function (CDF)** The Geometric RV X can take the values $\{1,\ 2,\ 3, \ldots\}$

$$\begin{aligned} \Pr(X = 1) &= p \\ \Pr(X = 2) &= (1-p)p \\ \Pr(X = 3) &= (1-p)^2 p \\ &\vdots \\ \Pr(X = k) &= (1-p)^{k-1} p \end{aligned}$$

  Therefore, the PMF of the Geometric RV is given by:

$$P_X(k) = (1-p)^{k-1} p$$

  and the CDF is given by

$$F_X(k) = 1 - (1-p)^k$$

- **Practical Applications:**
  - **Quality control:** Number of products inspected until the first defective item is found
  - **Call centers:** Number of calls until one is answered
  - **Maintenance:** Number of operational cycles until the first failure
  - **Clinical trials:** Number of patients treated until the first one recovers
  - **Networking:** Number of packet transmissions until the first is successfully received.
  - **Trading:** Number of customers enter a store until the first customer makes a purchase.

Example 3.4. At a "busy time," a call-center is very near capacity, so callers have difficulty placing their calls. It may be of interest to know the number of attempts necessary in order to make a connection. Suppose that we let $p = 0.2$ be the probability of a connection during a busy time. Find:

a) The probability that 6 attempts are necessary for a successful call.

b) From quality assurance perspective, assume that a call center is approved if the number of attempts until answering the customer should not exceed 4 times for 80% of the times. Does, the call-center considered in this example satisfy this quality assurance requirement?

c) If the system needs to comply with the quality assurance requirements, what is the minimum value of p that is required to achieve this.

Solution:

a) The number of attempts until the first successful call is modeled by a Geometric random variable with $p = 0.2$. Then,

$$\Pr(X = 6) = (1-p)^{6-1} p = 0.0655$$

b) The probability that the number of attempts until answering the customer does not exceed 4 times is given by

$$\Pr(X \leq 4) = 1 - (1-p)^4 = 0.5904$$

Therefore, the system does not satisfy the 80% requirement.

c) In order to comply with the quality assurance requirements, we need to increase the value of p to $\tilde{p}$ such that:

$$\Pr(X \leq 4) = 1 - (1-\tilde{p})^4 \geq 0.80$$

Therefore,

$$(1-\tilde{p})^4 \leq 0.2 \quad \Rightarrow \quad \tilde{p} \geq 1 - (0.2)^{1/4} \quad \Rightarrow \quad \tilde{p} \geq 0.3313.$$

Hence, the minimum value of the answering rate should be $p = 0.3313$ in order to be approved by the quality assurance agency.

3.5 Negative Binomial Random Variable

- **Description**
  The **Negative Binomial distribution** models the number of *trials* needed to achieve a fixed number of **successes** in a sequence of independent Bernoulli trials (i.e., each trial results in either success or failure).

  It is especially useful in modeling **waiting times** and **counting processes**, such as:

  - How many emails must you send before getting 5 replies?
  - How many free throws must a basketball player attempt before scoring 10 baskets?
  - How many inspections until 3 defective items are found?

Unlike the Binomial distribution, where the number of trials is fixed and we count the number of successes, the Negative Binomial fixes the number of successes and counts how many trials are needed.

The Negative Binomial RV is also called the **Generalized Geometric** random variable or **Pascal** random variable.

Definition 8. Let:

- p: the probability of success in each Bernoulli trial
- k: the target number of successes
- X: the number of trials needed to get k successes

Then X follows a **Negative Binomial distribution** with parameters k and p, denoted as:

$$X \sim \text{NegBin}(k, p)$$

The support of X is:

$$x = k, k+1, k+2, \ldots$$

- **Probability Mass Function (PMF)**
  The PMF of a negative binomial random variable X is given by:

$$P_X(n) = \Pr(X = n) = \binom{n-1}{k-1} p^k (1-p)^{n-k}, \quad n = k, k+1, k+2, \ldots$$

- **Derivation of the PMF for the Negative Binomial Distribution**

  We are interested in finding the probability that the kth success occurs exactly on the nth trial, i.e., $\Pr(X = n)$.

  Key Idea:

  To get the kth success on the nth trial:

  - Among the first $n-1$ trials, there must be exactly $k-1$ successes and $n-k$ failures.
  - The nth trial must be a success.

  Derivation Steps

  - **Number of ways** to arrange $k-1$ successes and $n-k$ failures in the first $k-1$ trials:

$$\binom{n-1}{k-1}$$

  - **Probability** of such an arrangement (using independence of trials):

$$p^{k-1}(1-p)^{n-k}$$

  - **Probability** that the nth trial is a success:

$$p$$

  - **Multiplying all components**, we obtain:

$$\Pr(X = n) = \binom{n-1}{k-1} p^{k-1}(1-p)^{n-k} \cdot p$$

  - **Simplifying**:

$$\Pr(X = n) = \binom{n-1}{k-1} p^k (1-p)^{n-k}$$

  This is valid for:

$$n = k, k+1, k+2, \ldots$$

- **Practical Applications:**

  The negative binomial distribution has wide applications, particularly in situations where we count the number of attempts before achieving a set number of successful outcomes. Examples include:

- **Marketing:** Number of customer calls until reaching 10 purchases
- **Manufacturing:** Number of items tested until detecting 5 defective units
- **Insurance:** Number of claims filed until a certain number of them are accepted
- **Biology:** Number of experiments until observing a certain number of gene expressions
- **Epidemiology:** Modeling the spread of infectious diseases (count of events until outbreak threshold)

Example 3.5. In an NBA (National Basketball Association) championship series, the team that wins four games out of seven is the winner. Suppose that teams A and B face each other in the championship games and that team A has probability 0.55 of winning a game over team B.

a) What is the probability that team A will win the series in 6 games?

b) What is the probability that team A will win the series?

Solution:

a) The probability that team A will win the series in 6 games is equivalent the probability that the **4th success (win)** occurs **exactly on the 6th trial (game)**. This setup is a classic Negative Binomial Distribution problem.

Remember: In the Negative Binomial distribution, we compute the probability that the $k-th$ success happens on the $n-th$ trial, where each trial is independent and has a fixed success probability p. Hence:

$$NegBin(6;4,0.55) = \binom{5}{3} 0.55^4 (1-0.55)^{6-4} = 0.1853. \tag{18}$$

b) We want the total probability that Team A wins the series, regardless of whether it's in Game 4, 5, 6, or 7.

This is different from Part a (which was only about winning in **exactly 6 games**). Here, we consider all possible ways Team A could win the best-of-7 series.

Below are the possible scenarios that team A will win the series:

Games Played	**Meaning**	Probability
4 games	A wins first 4 games7	$NegBin(4,4,0.55)$
5 games	A wins 4 games, and B wins 1	$NegBin(5,4,0.55)$
6 games	A wins 4 games, and B wins 2	$NegBin(6,4,0.55)$
7 games	A wins 4 games, and B wins 3	$NegBin(7,4,0.55)$

Therefore, we'll sum the probabilities that Team A wins in **exactly 4, 5, 6, or 7 games**. That is:

$$\begin{aligned}\Pr(winning) &= NegBin(4;4,0.55) + NegBin(5;4,0.55) \\ &\quad + NegBin(6;4,0.55) + NegBin(7;4,0.55) \\ &= 0.0915 + 0.1647 + 0.1853 + 0.1668 = 0.6083.\end{aligned}$$

3.6 Discrete Uniform Random Variable

- **Description**
  The **discrete uniform random variable** is a fundamental probability distribution that models a scenario where each of a finite number of outcomes is equally likely. It is commonly used in problems involving games of chance, random sampling, and systems with symmetry or fairness assumptions.

  For example, rolling a fair die, drawing a card at random from a shuffled deck, or selecting an item uniformly from a finite set can all be modeled using a discrete uniform distribution.

Definition 9. Let X be a discrete random variable that takes on a finite set of n values:

$$\{x_1, x_2, \ldots, x_n\}$$

If each outcome is equally likely, then X is called a **discrete uniform random variable**.

The values are typically consecutive integers, e.g., $x_k = a + k - 1$ for $k = 1, 2, \ldots, n$, where a is the minimum value and $b = a + n - 1$ is the maximum value.

- **Probability Mass Function (PMF)**
  The probability mass function of a discrete uniform random variable X defined over the integers $a, a+1, \ldots, b$ (where $n = b - a + 1$) is:

$$P_X(x) = \begin{cases} \frac{1}{n} = \frac{1}{b-a+1}, & \text{if } x = a, a+1, \ldots, b \\ 0, & \text{otherwise} \end{cases}$$

- **Cumulative Distribution Function (CDF)**
  The cumulative distribution function (CDF) of X is given by:

$$F_X(x) = \Pr(X \leq x) = \begin{cases} 0, & x < a \\ \frac{\lfloor x \rfloor - a + 1}{b-a+1}, & a \leq x \leq b \\ 1, & x > b \end{cases}$$

- **Practical Applications**
  - **Wireless Communications:** In frequency hopping systems, channels are often selected uniformly at random from a set of available frequencies to minimize interference and eavesdropping.
  - **Computer Networks:** Routing protocols or load balancers may assign packets to paths or servers uniformly to evenly distribute the traffic across available resources.
  - **Industrial Automation and Control:** In manufacturing lines, testing schedules or component selection for quality control sampling can be modeled with uniform distributions when all components or times are equally likely.
  - **Random Access in Communication Protocols:** For example, in ALOHA or CSMA-based networks, backoff times or retransmission slots are selected uniformly from a predefined set to avoid collisions.

- **Scheduling and Task Allocation:** In embedded systems and real-time control, tasks or processes might be dispatched uniformly at random for fairness in shared-resource environments.
- **Sensor Networks:** In distributed sensing, nodes may wake up or transmit at uniform random intervals to reduce collisions and conserve energy.
- **Industrial Engineering:** Uniform distributions are used in modeling arrival times, demand estimation, or machine settings when no prior knowledge of preferences is known.
- **Robotics and Motion Planning:** Path planning algorithms may discretize the environment and select potential moves uniformly for exploration-based planning.

4 Continuous Random Variables

- Continuous RVs take any real value.
- A continuous random variable has a probability of 0 of assuming *exactly* any of its values. Why?
  Consider a discrete random variable, X, with N values $\{x_i = i/N; i = 0,\ 1,\ \ldots,\ N-1\}$ and all values are equal probable. That is, the probability mass function of X is

$$\Pr(x_i) = f_X(x_i) = \frac{1}{N} \tag{19}$$

  In this case, regardless how large is N (i.e., the number of possible values of the RV), we can still assign a positive probability $\frac{1}{N}$ for each value of the X. Now, if we consider a continuous version of X by increasing N to ∞. In this case, X can take any value in the interval $[0, 1)$. As consequence to this limit, $N \to \infty$, is that

$$\Pr(x_i) = f_X(x_i) = \lim_{N\to\infty} \frac{1}{N} = 0 \tag{20}$$

- Even though each point has zero probability of occurring in continuous RVs. Yet, something has to occur! This problem is common to continuous random variables, and it is clear that the probability mass function is not a suitable description for such a random variable. Alternatively, there are two descriptions to handle continuous RVs:
  - Cumulative Distribution Function (CDF)
  - Probability Density Function (PDF)

4.1 Cumulative Distribution Function (CDF)

Definition 10. The cumulative distribution function $F(x)$ of a continuous random variable X

$$F_X(x) = \Pr(X \leq x) \tag{21}$$

Properties of CDF

$$F_X(-\infty) = 0. \tag{22a}$$
$$F_X(\infty) = 1. \tag{22b}$$
$$0 \leq F_X(x) \leq 1. \tag{22c}$$
$$\text{if } x_1 < x_2, \text{ then } F_X(x_1) \leq F_X(x_2). \tag{22d}$$
$$P(x_1 < X \leq x_2) = F_X(x_2) - F_X(x_1). \tag{22e}$$
$$F_X(x) \text{ is continuous from right. That is } F_X(x) = F_X(x^+). \tag{22f}$$

Note 1. Property (22)-d indicates that the CDF is a monotonic non-decreasing function.

Note 2. CDFs are defined not only for the values assumed by the given random variable but for all real numbers.

Example 4.1 (Checking Whether a Function is a Valid CDF)**.** To verify that a function $F(x)$ is a valid cumulative distribution function (CDF), we usually check the following properties:

(1) $F_X(x) \in [0,\ 1]$

(2) $F(x)$ is **non–decreasing**.

(3) $\lim_{x \to -\infty} F(x) = 0.$

(4) $\lim_{x \to +\infty} F(x) = 1.$

(5) $F(x)$ is **right–continuous**.

Below are two examples: one valid CDF and one that is *not* a valid CDF.

Part 1: A Valid CDF Consider

$$F_X(x) = \begin{cases} 0, & x < 0, \\ \dfrac{x^2}{4}, & 0 \leq x < 2, \\ 1, & x \geq 2. \end{cases}$$

Check the properties.

- **Range.** The values of $F_X(x)$ lie between 0 and 1.
- **Monotonicity** For $x < 0$, $F_X(x) = 0$. For $0 \leq x < 2$, $F_X(x) = x^2/4$, which is an increasing function in x on $[0, 2)$. For $x \geq 2$, $F_X(x) = 1$, which is constant. Thus, $F_X(x)$ is *non–decreasing* on $\mathbb{R}$.

- **Limits at $\pm\infty$.**
  As $x \to -\infty$, we are in the region $x < 0$, hence
  $$\lim_{x\to-\infty} F_X(x) = \lim_{x\to-\infty} 0 = 0.$$
  As $x \to +\infty$, we are eventually in the region $x \geq 2$, where $F_X(x) = 1$. Thus,
  $$\lim_{x\to+\infty} F_X(x) = \lim_{x\to-\infty} 1 = 1.$$
- **Right–continuity**
  at $x = 0$,
  $$\lim_{x\downarrow 0} F_X(x) = \frac{0^2}{4} = 0 = F_X(0);$$
  at $x = 2$,
  $$\lim_{x\downarrow 2} F_X(x) = \frac{2^2}{4} = 1 = F_X(2).$$
  Between these points, $F_X(x)$ is continuous, and it is constant for $x \geq 2$. Hence, F_X is right–continuous everywhere.

Conclusion. All CDF properties are satisfied, so $F_X(x)$ is a **valid CDF**.

Part 2: A Function That Is *Not* a Valid CDF Consider the function
$$G_X(x) = \frac{1}{2}\big(1 + \sin x\big), \qquad x \in \mathbb{R}.$$

Check the properties.

- **Range.** Since $-1 \leq \sin x \leq 1$, we have
  $$0 \leq G_X(x) = \tfrac{1}{2}(1 + \sin x) \leq 1,$$
  so the values of $G_X(x)$ lie between 0 and 1. This is necessary, but not sufficient, for a CDF.
- **Monotonicity.** A CDF must be *non–decreasing*. However, $G_X(x)$ oscillates because of the sine term. Its derivative is
  $$G'_X(x) = \frac{1}{2}\cos x,$$
  which is positive on some intervals (e.g., around $x = 0$) and negative on others (e.g., around $x = \pi$). Therefore, $G_X(x)$ increases and decreases repeatedly and is **not** non–decreasing.
- **Limits at $\pm\infty$.** As $x \to \infty$ or $x \to -\infty$, the function $\sin x$ keeps oscillating between -1 and 1. Hence, $G_X(x)$ does not converge to a single limit. In particular,
  $$\lim_{x\to\infty} G_X(x) \text{ does not exist}, \qquad \lim_{x\to-\infty} G_X(x) \text{ does not exist},$$
  so we certainly do not have the required $\lim_{x\to-\infty} G_X(x) = 0$ and $\lim_{x\to+\infty} G_X(x) = 1$.

Conclusion. Although $G_X(x)$ always takes values between 0 and 1, it is **not** a valid CDF because:

1. it is not non–decreasing (it oscillates up and down), and
2. it does not converge to 0 as $x \to -\infty$ nor to 1 as $x \to +\infty$.

Note: If any of the required conditions is violated, we can immediately conclude that the function is not a valid CDF, and there is no need to check the remaining conditions. In the above example, even though the function already fails the monotonicity condition, we also examined the limits as an illustration, since some of you might begin by checking the limit conditions first.

Example 4.2. Which of the following mathematical functions could be the CDF of some random variable?

A) $F_X(x) = \frac{1}{2} + \frac{1}{\pi}\tan^{-1}(x)$

B) $F_X(x) = [1 - \mathrm{e}^{-x}]\, u(x)$, $\quad (u(x)$ is the unit step function).

C) $F_X(x) = \mathrm{e}^{-x^2}$

D) $F_X(x) = x^2 u(x)$

Solution:
To determine this, we need to check that the function starts at zero when $x = -\infty$, ends at one when $x = \infty$, and is monotonic increasing in between. The first two functions satisfy these properties and thus are valid CDFs while the last two do not. The function in (c) is decreasing for positive values of x, while the function in (d) takes on values greater than 1 and $F_X(\infty) \neq 1$

Example 4.3 (Computing Probabilities using the CDF)**.** Suppose a random variable has a CDF given by $F_X(x) = 1 - e^{-x}$. Find the following probabilities:

a) $\Pr(X < 5)$

b) $\Pr(X > 5)$

c) $\Pr(3 < X < 7)$

d) $\Pr(X > 5 | X < 7)$

e) $\Pr(2X < 5)$

Solution:

a) Note that $F_X(5)$ gives us $\Pr(X \leq 5)$, which is not quite what we want. However, we note that

$$F_X(5) = \Pr(\{X < 5\} \cup \{X = 5\}) = \Pr(X < 5) + \Pr(X = 5)$$

Hence,

$$\Pr(X < 5) = F_X(5) - \Pr(X = 5)$$

In this case, since X is a continuous random variable, $\Pr(X = 5) = 0$ and so there is no need to make a distinction between $\Pr(X < 5)$ and $\Pr(X \leq 5)$; however, for discrete random variables, we would need to be careful. Accordingly, $\Pr(X < 5) = F_X(5) = 1 - \exp(-5)$.

b) $\Pr(X > 5) = 1 - \Pr(X \leq 5) = 1 - F_X(5) = \mathrm{e}^{-5}$

c) We note that in general $F_X(7) - F_X(3) = \Pr(3 < X \leq 7)$. Again, for this continuous random variable, $\Pr(X = 7) = 0$, so we can also write $\Pr(3 < X < 7) = F_X(7) - F_X(3) = \mathrm{e}^{-3} - \mathrm{e}^{-7}$.

d) For part (d) we invoke the definition of conditional probability to write the required quantity in terms of the CDF of X :

$$\begin{aligned} \Pr(X > 5 \mid X < 7) = \frac{\Pr(\{X > 5\} \cap \{X < 7\})}{\Pr(X < 7)} = \frac{\Pr(5 < X < 7)}{\Pr(X < 7)} &= \frac{F_X(7) - F_X(5)}{F_X(7)} \\ &= \frac{\mathrm{e}^{-5} - \mathrm{e}^{-7}}{1 - \mathrm{e}^{-7}} \end{aligned} \tag{23}$$

e) $\Pr(2X < 5) = \Pr(X < 5/2) = F_X(5/2) = 1 - e^{-2.5}$

4.2 Probability Density Function (PDF)

- Motivation to work with the probability density function:
  - While the CDF introduced in the last section represents a mathematical tool to statistically describe a random variable, it is often quite cumbersome to work with CDFs. For example, we will see later in this chapter that the most important and commonly used random variable, the Gaussian random variable, has a CDF which cannot be expressed in closed form.
  - Furthermore, it can often be difficult to infer various properties of a random variable from its CDF.
- Definition:

Definition 11. The PDF of the random variable X evaluated at the point x is

$$f_X(x) = \lim_{\varepsilon \to 0} \frac{\Pr(x \leq X < x + \varepsilon)}{\varepsilon} \tag{24}$$

As the name implies, the PDF is the probability that the random variable X lies in an infinitesimal interval about the point $X = x$, normalized by the length of the interval

- Relation between PDF and CDF:
  For continuous random variables,

$$\Pr(x \leq X < x + \varepsilon) = F_X(x + \varepsilon) - F_X(x)$$

so that

$$f_X(x) = \lim_{\varepsilon \to 0} \frac{F_X(x + \varepsilon) - F_X(x)}{\varepsilon} = \frac{\mathrm{d}F_X(x)}{\mathrm{d}x} \tag{25}$$

Hence, the PDF of a random variable is the derivative of its CDF. Conversely, the CDF of a random variable can be expressed as the integral of its PDF. This property is illustrated in Figure 5.

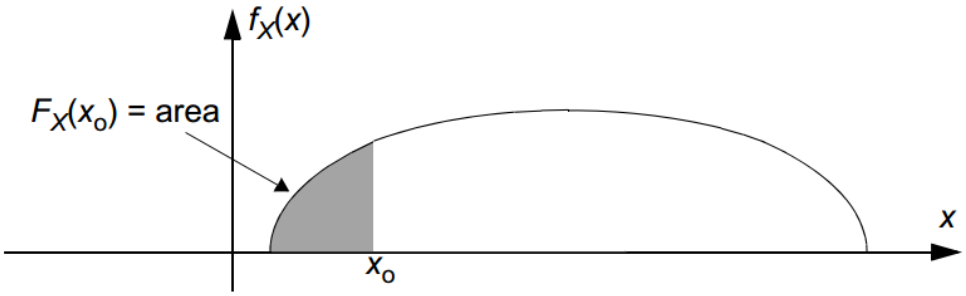


Figure 5: Relationship between the PDF and CDF of a random variable.

- Properties of the PDF:

$$(1) \quad f_X(x) \geq 0 \tag{26}$$

$$(2) \quad f_X(x) = \frac{\mathrm{d}F_X(x)}{\mathrm{d}x} \tag{27}$$

$$(3) \quad F_X(x) = \int_{-\infty}^{x} f_X(y)\mathrm{d}y \tag{28}$$

$$(4) \quad \int_{-\infty}^{\infty} f_X(x)\mathrm{d}x = 1 \tag{29}$$

$$(5) \quad \int_{a}^{b} f_X(x)\mathrm{d}x = \Pr(a < X \leq b) \tag{30}$$

Note 3. The PDF it is not restricted to be less than unity as with the CDF. In other words, $f_X(x)$ can take values greater than 1.

Example 4.4 (Verifying a Valid PDF)**.** Consider the function

$$f_X(x) = \begin{cases} kx, & 0 \leq x \leq 2, \\ 0, & \text{otherwise}, \end{cases}$$

where k is a constant. We want to check whether $f_X(x)$ can be a valid probability density function (PDF), and if so, find the value of k.

Solution:

Condition 1: Non–negativity. For $0 \le x \le 2$, we have $x \ge 0$. If $k \ge 0$, then

$$f_X(x) = kx \ge 0.$$

Outside the interval $[0, 2]$, we have $f_X(x) = 0$. Thus, as long as $k \ge 0$, the non–negativity condition is satisfied:

$$f_X(x) \ge 0 \quad \text{for all } x \in \mathbb{R}.$$

Condition 2: Total area equals 1. A valid PDF must satisfy

$$\int_{-\infty}^{\infty} f_X(x)\,dx = 1.$$

Here,

$$\int_{-\infty}^{\infty} f_X(x)\,dx = \int_0^2 kx\,dx = k\int_0^2 x\,dx = k\left[\frac{x^2}{2}\right]_0^2 = k\left(\frac{2^2}{2} - 0\right) = 2k.$$

For this to be a PDF, we require

$$2k = 1 \quad \Longrightarrow \quad k = \frac{1}{2}.$$

Conclusion. With $k = \frac{1}{2}$, the function

$$f_X(x) = \begin{cases} \frac{1}{2}x, & 0 \le x \le 2, \\ 0, & \text{otherwise}, \end{cases}$$

is a valid PDF.

Example 4.5 (Verifying a Valid PDF)**.** Consider the function

$$h_X(x) = \begin{cases} 4x, & 0 \le x \le 1, \\ 0, & \text{otherwise}. \end{cases}$$

Verify whether $f_X(x)$ can be a valid probability density function (PDF) or not.

Solution:

Non–negativity. Clearly,

$$h_X(x) = 4x \geq 0 \quad \text{for } 0 \leq x \leq 1,$$

and $h_X(x) = 0$ otherwise, so non–negativity holds.

Total area (integral). Check the integral:

$$\int_{-\infty}^{\infty} h_X(x)\,dx = \int_0^1 4x\,dx = 4\int_0^1 x\,dx = 4\left[\frac{x^2}{2}\right]_0^1 = 4\cdot\frac{1}{2} = 2 \neq 1.$$

Conclusion. The function $h_X(x)$ is non–negative, but its total area is 2, not 1. Hence, $h_X(x)$ is *not* a valid probability density function. To turn it into a valid PDF, we would need to normalize it by dividing by 2, i.e., use $\tilde{h}_X(x) = \frac{1}{2}h_X(x) = 2x$ on $[0,1]$, which matches the previous valid example.

Example 4.6. Which of the following are valid PDFs?

A) $f_X(x) = \mathrm{e}^{-x}u(x)$

B) $f_X(x) = \mathrm{e}^{-|x|}$

C) $f_X(x) = \begin{cases} \frac{3}{4}\left(x^2-1\right), & |x| < 2 \\ 0, & \text{otherwise} \end{cases}$

D) $f_X(x) = \begin{cases} 1, & 0 \leq x < 1 \\ 0, & \text{otherwise} \end{cases}$

E) $f_X(x) = 2x\mathrm{e}^{-x^2}u(x)$

Solution: To verify the validity of a potential PDF, we need to verify only that the function is nonnegative and the area underneath the function is equal to unity. The function in part (c) takes on negative values, while the function in part (b) is not properly normalized, and therefore these are not valid PDFs. The other three functions are valid PDFs.

Example 4.7. A random variable has a CDF given by $F_X(x) = \left(1 - \mathrm{e}^{-\lambda r}\right)u(x)$. Find its PDF?
Solution: its PDF is given by

$$f_X(x) = \frac{\mathrm{d}F_X(x)}{\mathrm{d}x} = \lambda\mathrm{e}^{-\lambda x}u(x)$$

Example 4.8. If a random variable has a PDF given by $f_X(x) = 2x\mathrm{c}^{-x^2}u(x)$, find its CDF?
Solution: its CDF is given by

$$F_X(x) = \int_{-\infty}^{x} f_X(y)\mathrm{d}y = \int_{-\infty}^{x} 2y\mathrm{e}^{-y^2}u(y)\mathrm{d}y = \int_{0}^{x} 2y\mathrm{e}^{-y^2}\mathrm{d}yu(x) = \left(1 - \mathrm{e}^{-x^2}\right)u(x)$$

Example 4.9 (Example: Using the PDF (and CDF) to Compute a Probability)**.** Consider a network server whose *time to failure* (in hours) is modeled by an exponential random variable X with rate $\lambda = 0.5$ failures per hour. That is,

$$X \sim \mathrm{Exp}(\lambda = 0.5).$$

The probability density function (PDF) of X is

$$f_X(x) = \begin{cases} 0.5\, e^{-0.5x}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

What is the probability that the server fails between 1 and 3 hours from now?

Solution: The probability that the server fails between 1 and 3 hours from now is equivalent to

$$Pr(1 < X < 3).$$

We will solve this in two equivalent ways:

1. Directly from the PDF by integration.
2. Using the CDF obtained from the PDF.

Using the PDF By definition,

$$P(1 < X < 3) = \int_1^3 f_X(x)\, dx = \int_1^3 0.5\, e^{-0.5x}\, dx.$$

Compute the integral:

$$\int 0.5\, e^{-0.5x}\, dx = 0.5 \cdot \left(\frac{-2}{e^{0.5x}}\right) = -e^{-0.5x} + C.$$

Thus,

$$P(1 < X < 3) = \left[-e^{-0.5x}\right]_{x=1}^{x=3} = \left(-e^{-0.5\cdot 3}\right) - \left(-e^{-0.5\cdot 1}\right) = e^{-0.5} - e^{-1.5}.$$

So,

$$P(1 < X < 3) = e^{-0.5} - e^{-1.5}.$$

Using the CDF First, we obtain the cumulative distribution function (CDF) from the PDF. For $x \geq 0$,

$$F_X(x) = P(X \leq x) = \int_0^x 0.5\, e^{-0.5t}\, dt.$$

As computed above,

$$\int 0.5\, e^{-0.5t}\, dt = -e^{-0.5t} + C,$$

so

$$F_X(x) = \left[-e^{-0.5t}\right]_{t=0}^{t=x} = -e^{-0.5x} - \left(-e^{0}\right) = 1 - e^{-0.5x}, \quad x \geq 0.$$

Now,

$$P(1 < X < 3) = F_X(3) - F_X(1).$$

Compute each term:

$$F_X(3) = 1 - e^{-0.5 \cdot 3} = 1 - e^{-1.5},$$
$$F_X(1) = 1 - e^{-0.5 \cdot 1} = 1 - e^{-0.5}.$$

Therefore,

$$P(1 < X < 3) = F_X(3) - F_X(1) = \left(1 - e^{-1.5}\right) - \left(1 - e^{-0.5}\right) = e^{-0.5} - e^{-1.5}.$$

Conclusion Both methods give the same result:

$$P(1 < X < 3) = e^{-0.5} - e^{-1.5}.$$

This example shows how we can:

- compute a probability directly from the PDF by integration, and
- obtain the CDF from the PDF and then use $P(a < X < b) = F_X(b) - F_X(a)$ to get the same probability.

5 The Gaussian (or Normal) Random Variable

5.1 Motivation to study the Gaussian random variable

Many real–life quantities are well modeled by a Gaussian (normal) random variable:

- Heights of people, Exam scores
- Measurement noise in communication systems
- Thermal noise in resistors, etc.

Because of the **Central Limit Theorem**, sums of many small random effects tend to look Gaussian, which is why this distribution appears everywhere.

5.2 PDF of the Gaussian random variable

Definition 12 (Normal/Gaussian RV)**.** A *normal (Gaussian) random variable* X with mean μ and standard deviation σ is written as

$$X \sim \mathcal{N}(\mu, \sigma^2).$$

Its probability density function (pdf) is

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad -\infty < x < \infty.$$

Shape and Role of Parameters An example of a Guassian PDF is shown in Figure 6. The graph of $f_X(x)$ is the well–known *bell curve*:

- It is *symmetric* around $x = \mu$.
- It has a single peak at $x = \mu$.
- The tails approach zero as $x \to \pm\infty$.

Effect of the parameters:

- The Gaussian random variable has two parameters:
  - Mean (μ).
  - Variance (σ^2)
- Changing μ shifts the curve left or right (horizontal translation).
- Changing σ stretches or compresses the curve:
  - small σ: narrow and tall (values tightly clustered around μ),
  - large σ: wide and flatter (values more spread out).

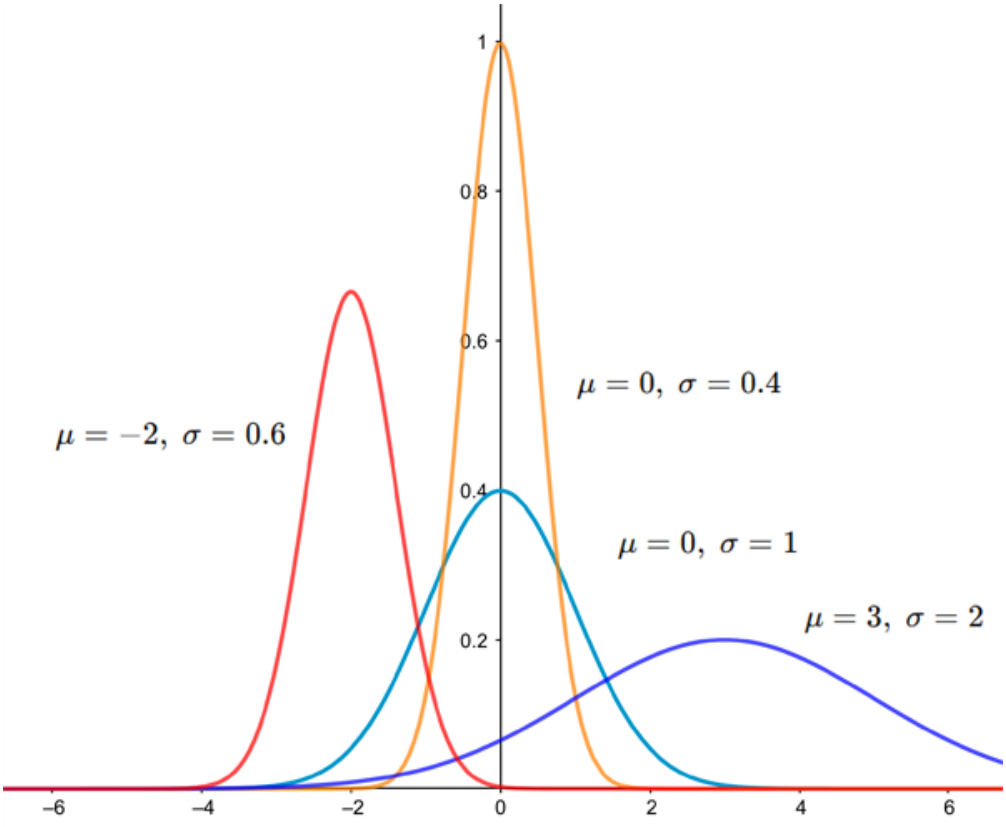


Figure 6: PDF of a Gaussian random variable for various mean and standard deviation values.

The 68–95–99.7 Rule

For any normal distribution:

$$P(\mu - \sigma \leq X \leq \mu + \sigma) \approx 0.68,$$
$$P(\mu - 2\sigma \leq X \leq \mu + 2\sigma) \approx 0.95,$$
$$P(\mu - 3\sigma \leq X \leq \mu + 3\sigma) \approx 0.997.$$

So about 68%, 95%, and 99.7% of values lie within 1, 2, and 3 standard deviations of the mean, respectively.

5.3 CDF of the Gaussian random variable

- The CDF of a Gaussian random variable is written as

$$F_X(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-\mu)^2}{2\sigma^2}\right) \mathrm{d}y \tag{31}$$

  It can be shown that it is impossible to express this integral in closed form. However, there are two methods to dealing with this problem:

  – Numerical tables

  - Mathematical software packages.

- Whether the Gaussian CDF is to be evaluated by using a table or a program, the required CDF must be converted into one of a few commonly used standard forms. A few of these common forms are
  - error function integral, $\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x \exp\left(-t^2\right) \mathrm{d}t$
  - complementary error function integral, $\operatorname{erfe}(x) = 1 - \operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty \exp\left(-t^2\right) \mathrm{d}t$
  - Φ -function, $\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x \exp\left(-\frac{t^2}{2}\right) \mathrm{d}t$
  - Q -function, $Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty \exp\left(-\frac{t^2}{2}\right) \mathrm{d}t$

- Evaluating the CDF of a general Gaussian RV in terms of Φ-function:
  Starting with the Gaussian CDF in Equation (31), make the transformation $t = (y - \mu)/\sigma$, resulting in

$$F_X(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-\mu)^2}{2\sigma^2}\right) \mathrm{d}y = \int_{-\infty}^{\frac{x-\mu}{\sigma}} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right) \mathrm{d}t = \Phi\left(\frac{x-\mu}{\sigma}\right) \quad (32)$$

  Hence,

$$\Pr(X \leq x) = F_X(x) = \Phi\left(\frac{x-\mu}{\sigma}\right) \quad (33)$$

  That is to evaluate the CDF of a Gaussian random variable, we just evaluate the Φ -function at the points $(x - \mu)/\sigma$.

- Evaluating the CDF of a general Gaussian RV in terms of Q-function:
  The Q -function is more natural for cvaluating probabilitics of the form $\Pr(X > x)$. Following a line of reasoning identical to the previous paragraph, it is seen that if $X \sim N\left(m, \sigma^2\right)$, then

$$\Pr(X > x) = \int_{\frac{x-m}{\sigma}}^\infty \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right) \mathrm{d}t = Q\left(\frac{x-\mu}{\sigma}\right) \quad (34)$$

  Hence,

$$\Pr(X \leq x) = F_X(x) = 1 - Q\left(\frac{x-\mu}{\sigma}\right) \quad (35)$$

- Given that $X \sim N\left(m, \sigma^2\right)$, the following summarizes the relations between $F_X(x)$, Q, Φ, erf and erfc functions

$$F_X(x) = 1 - Q\left(\frac{x-\mu}{\sigma}\right) \tag{36}$$

$$F_X(x) = \Phi\left(\frac{x-\mu}{\sigma}\right) \tag{37}$$

$$Q(x) = 1 - \Phi(x) \tag{38}$$

$$\Phi(x) = \frac{1}{2}\left(1 + \text{erf}\left(\frac{x}{\sqrt{2}}\right)\right) = 1 - \frac{1}{2}\,\text{erfc}\left(\frac{x}{\sqrt{2}}\right) \tag{39}$$

$$\text{erf}(x) = 2\Phi(\sqrt{2}x) - 1 \tag{40}$$

Some of these relationships can be visualized using the graphical definitions of the Φ-function (phi function) and the Q-function shown in Figure 7.

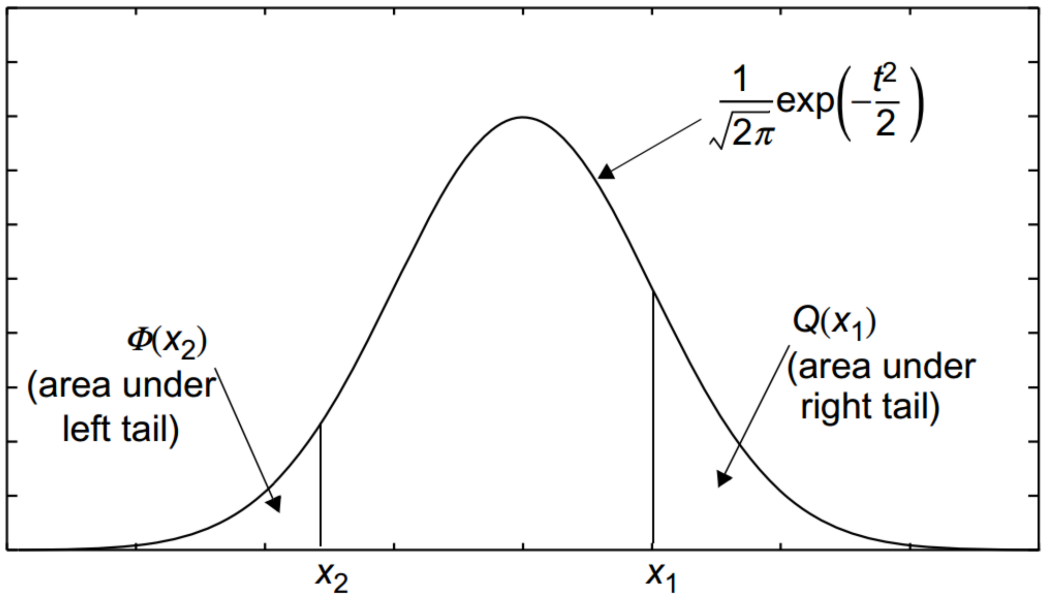


Figure 7: Standardized integrals related to the Gaussian CDF: the $\Phi(\cdot)$ and $Q(\cdot)$ functions.

Note 4. It should be noted that some internally defined programs for Q and related functions may expect a positive argument. If it is required to evaluate the Q -function at a negative value, the relationship $Q(x) = 1 - Q(-x)$ can be used. That is, to evaluate $Q(-2)$, for example, $Q(2)$ can be evaluated and then use $Q(-2) = 1 - Q(2)$.

Example 5.1. A random variable has a PDF given by

$$f_X(x) = \frac{1}{\sqrt{8\pi}} \exp\left(-\frac{(x+3)^2}{8}\right)$$

Find each of the following probabilities and express the answers in terms of Φ and Q-functions.

A) $\Pr(X \leq 0)$

B) $\Pr(X > 4)$

C) $\Pr(|X + 3| < 2)$

Solutions: For the given Gaussian PDF, $\mu = -3$ and $\sigma = 2$.

A) $\Pr(X \leq 0) = \Phi((0 - (-3))/2) = \Phi(1.5)$. This can be rewritten in terms of a Q-function as $\Pr(X \leq 0) = 1 - Q(1.5)$

B) The probability in part (B) is easier to express directly in terms of a Q-function. Hence, $\Pr(X > 4) = Q((4-) - 3))/2) = Q(3.5)$

C) The probability considered here is an interval. That is,

$$\begin{aligned}\Pr(|X + 3| < 2) &= \Pr(-5 < X < -1) = F_X(X < -1) - F_X(X < -5) \\ &= \Phi((-1 - (-3))/2) - \Phi((-5 - (-3))/2) = \Phi(1) - \Phi(-1) \\ &= \Phi(1) - (1 - \Phi(1)) = 2\Phi(1) - 1\end{aligned}$$

In terms of Q-function,

$$\Pr(|X + 3| < 2) = 2(1 - Q(x)) - 1 = 1 - 2Q(x)$$

5.4 The Standard Normal Distribution

5.4.1 Definition

The *standard normal distribution* is a special normal distribution with

$$\mu = 0, \quad \sigma = 1.$$

We denote

$$Z \sim \mathcal{N}(0, 1).$$

Its pdf is

$$\phi(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right), \quad -\infty < z < \infty.$$

The cumulative distribution function (cdf) is

$$\Phi(z) = P(Z \leq z).$$

There is no simple closed-form expression for $\Phi(z)$, so in practice we use:

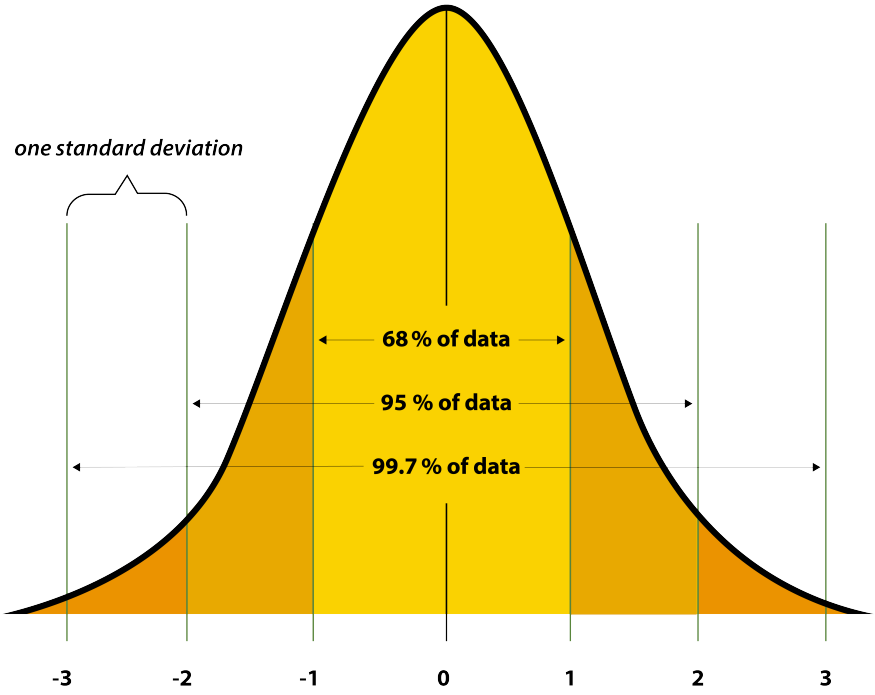


Figure 8: Plot of the standard normal distribution.

- standard normal tables (Z-tables),
- calculators or software.

5.4.2 Standardizing a Normal Random Variable

If $X \sim \mathcal{N}(\mu, \sigma^2)$, we can convert it to a standard normal variable Z by

$$Z = \frac{X - \mu}{\sigma}.$$

This process is called *standardization*. Then $Z \sim \mathcal{N}(0, 1)$, and we can use the standard normal table to compute probabilities.

Example 5.2 (Standardization)**.** If $X \sim \mathcal{N}(70, 10^2)$ and $x = 85$, then

$$z = \frac{85 - 70}{10} = 1.5.$$

So the value $x = 85$ in the original scale corresponds to $z = 1.5$ in the standard normal scale.

5.4.3 Basic Properties of the Standard Normal

1. Symmetry. The standard normal distribution is symmetric about zero:

$$P(Z \le -z) = P(Z \ge z) \quad \longleftrightarrow \quad \Phi(-z) = 1 - \Phi(z)$$
$$P(Z \ge -z) = P(Z \le z) \quad \longleftrightarrow \quad 1 - \Phi(-z) = \Phi(z)$$

2. Mean and variance.

$$E[Z] = 0, \qquad \mathrm{Var}(Z) = 1.$$

3. The 68–95–99.7 rule. For any normal distribution (including the standard normal),

$$P(-1 < Z < 1) \approx 0.68,$$
$$P(-2 < Z < 2) \approx 0.95,$$
$$P(-3 < Z < 3) \approx 0.997.$$

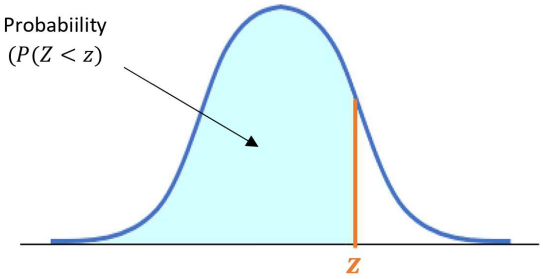
That is, about 68%, 95%, and 99.7% of the values lie within 1, 2, and 3 standard deviations of the mean, respectively as shown in Figure 8.

5.4.4 Using the Standard Normal Table (also called Z-Table)

The standard normal table (Z–table) provides values of the cumulative distribution function $\Phi(z) = P(Z \le z)$ for the standard normal random variable $Z \sim \mathcal{N}(0, 1)$. Since $\Phi(z)$ has no simple closed–form expression, these tabulated values are essential for computing probabilities and critical values in problems involving Gaussian random variables.

An example of the "Standard Normal Table" is shown in Figure 9.

- Defining the probability in terms of $\Phi(z)$.
  Most tables give $\Phi(z) = P(Z \le z)$ for $z \ge 0$.
  - **Probabilities of the Form** $P(Z \le z)$
    - If $z \ge 0$, read $\Phi(z)$ directly from the table.
    - If $z < 0$, use symmetry:
      $$P(Z \le z) = 1 - \Phi(-z).$$
  - **Probabilities of the Form** $P(Z \ge z)$
    $$P(Z \ge z) = 1 - P(Z \le z) = 1 - \Phi(z).$$
  - **Probabilities Between Two Values**
    For $a < b$,
    $$P(a < Z < b) = \Phi(b) - \Phi(a).$$



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5754
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7258	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7518	0.7549
0.7	0.7580	0.7612	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7996	0.8023	0.8051	0.8079	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9430	0.9441
1.6	0.9452	0.9463	0.9474	0.9485	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9700	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9762	0.9767
2.0	0.9773	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9865	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9980	0.9980	0.9981
2.9	0.9981	0.9982	0.9983	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998	0.9998

Figure 9: Standard Normal Table (Z-Table).

- If you start from a non-standard normal variable X, you first standardize to Z.

Example 5.3 (Using Z-Table with Standard Normal Distribution)**.** Let $Z \sim \mathcal{N}(0,1)$, find the following probabilities:

a) $Pr(Z \leq 1.2)$
Solution: From the table,

$$\Phi(1.2) \approx 0.8849.$$

Thus,

$$P(Z \leq 1.2) \approx 0.8849.$$

b) $Pr(Z \geq 1.2)$
Solution:

$$P(Z \geq 1.2) = 1 - \Phi(1.2) \approx 1 - 0.8849 = 0.1151.$$

So the probability is about 11.51%.

c) $Pr(-1 < Z < 2)$
Solution:

$$P(-1 < Z < 2) = \Phi(2) - \Phi(-1).$$

Using symmetry,

$$\Phi(-1) = 1 - \Phi(1).$$

Suppose from the table:

$$\Phi(2) \approx 0.9772, \qquad \Phi(1) \approx 0.8413.$$

Then

$$\Phi(-1) = 1 - 0.8413 = 0.1587,$$

and

$$P(-1 < Z < 2) = 0.9772 - 0.1587 = 0.8185.$$

So about 81.85% of the values lie between -1 and 2.

Example 5.4 (Using Z-Table with General Normal Distribution)**.** Let $X \sim \mathcal{N}(100, 15^2)$. Find $Pr(X < 115)$.

1. Standardize:
$$Z = \frac{X - 100}{15}, \quad z = \frac{115 - 100}{15} = \frac{15}{15} = 1.$$
2. Use the table:
$$P(X < 115) = P(Z < 1) = \Phi(1) \approx 0.8413.$$

Thus, there is about an 84.13% probability that $X < 115$.

Example 5.5 (Probability for a Gaussian Random Variable Using Tables)**.** A machine produces metal rods. The length X (in cm) of each rod is modeled as a Gaussian (normal) random variable:

$$X \sim \mathcal{N}(\mu = 10,\ \sigma^2 = 0.3^2),$$

where

$$\mu = 10 \text{ cm}, \qquad \sigma = 0.3 \text{ cm}.$$

What is the probability that a randomly chosen rod is shorter than 9.7 cm? That is, compute

$$P(X < 9.7).$$

Solution:
Step 1: Standardize (convert to Z)

Define the standard normal random variable

$$Z = \frac{X - \mu}{\sigma} = \frac{X - 10}{0.3}.$$

For the value $X = 9.7$, we obtain

$$z = \frac{9.7 - 10}{0.3} = \frac{-0.3}{0.3} = -1.0.$$

Thus,

$$P(X < 9.7) = P(Z < -1.0)\,.$$

Step 2: Use the standard normal distribution

Using the symmetry of the standard normal distribution, we have

$$P(Z < -1.0) = 1 - P(Z \leq 1.0).$$

From the standard normal CDF (looked up in a table), we find

$$\Phi(1.0) = P(Z \leq 1.0) \approx 0.8413.$$

Therefore,

$$P(Z < -1.0) = 1 - 0.8413 = 0.1587.$$

Final Answer

$$P(X < 9.7) \approx 0.1587.$$

This means that there is approximately a 15.87% chance that a randomly chosen rod is shorter than 9.7 cm.

5.4.5 Inverse Use: Finding z for a Given Probability

Sometimes we know a probability and want the corresponding z-value.

Example. Find z such that

$$P(Z \leq z) = 0.95.$$

We look in the table for the probability closest to 0.95 and find the corresponding z. This value is approximately

$$z \approx 1.645.$$

This z-value is called a *quantile* or *critical value* of the standard normal distribution. Such values are widely used in:

- constructing confidence intervals,
- hypothesis testing,
- defining thresholds in quality control and other applications.

Example 5.6 (Inverse of a Gaussian Random Variable Using Tables)**.** Consider a national math exam where the scores are modeled as a Gaussian (normal) random variable:

$$X \sim \mathcal{N}(\mu, \sigma^2), \quad \mu = 70,\ \sigma = 10.$$

The top 10% of students receive an "A" grade.

What is the minimum score x_A a student must obtain to be in the top 10%? Use the standard normal table to solve.

Solution:

Step 1: Translate the question into a probability statement

Being in the top 10% means:

$$P(X \geq x_A) = 0.10.$$

Equivalently,

$$P(X \leq x_A) = 0.90.$$

Thus, we want x_A such that

$$F_X(x_A) = 0.90,$$

i.e., x_A is the 90-th percentile of the distribution of X.

Step 2: Standardize to the standard normal

Define the standard normal random variable

$$Z = \frac{X - \mu}{\sigma} = \frac{X - 70}{10}.$$

Then

$$P(X \leq x_A) = 0.90 \quad \Longrightarrow \quad P\left(Z \leq z_{0.90}\right) = 0.90,$$

where $z_{0.90}$ is the 90-th percentile of the standard normal distribution.

Step 3: Use the standard normal table (inverse step)

From the standard normal CDF table, we search for the probability value closest to 0.9000. This typically corresponds to

$$z_{0.90} \approx 1.28.$$

Step 4: Convert back to the original variable X

We have

$$z_{0.90} = \frac{x_A - 70}{10}.$$

Hence,

$$x_A = 70 + 10 \times 1.28 = 70 + 12.8 = 82.8.$$

Final Answer

The minimum score required to obtain an "A" is approximately

$$x_A \approx 83 \text{ marks}.$$

This procedure illustrates the use of the *inverse* of the Gaussian CDF (also called the quantile function): we start from a probability value (here 0.90), find the corresponding z-value from the standard normal table, and then map this z-value back to the original random variable X.

5.4.6 Short Summary

- The *standard normal distribution* is a normal distribution with mean 0 and variance 1.
- Any normal $X \sim \mathcal{N}(\mu, \sigma^2)$ can be converted to standard normal using

$$Z = \frac{X - \mu}{\sigma}.$$

- Probabilities involving normal variables are usually computed by:
  1. standardizing to Z,
  2. using the standard normal table $\Phi(z)$ together with symmetry properties.

6 Common Continuous Distributions

Common continuous random variables (RVs) provide standard models for many real-life phenomena and are used repeatedly in probability and statistics. Rather than inventing a new distribution for every problem, we rely on a small family of well-understood models. For example, the uniform RV describes outcomes that are equally likely over an interval, the exponential RV is widely used for modeling waiting times between events, and the Gaussian (normal) RV arises naturally in situations involving many small random effects, such as measurement noise. Other important continuous RVs, such as the gamma, Rayleigh, and chi-square distributions, appear in reliability analysis, wireless communications, and hypothesis testing.

In the following sections, we will introduce these common continuous RVs, their probability density functions (pdfs) and CDFs, and discuss when each model is most appropriate to use. For each RV, the key to computing probabilities is its cumulative distribution function (CDF) $F_X(x)$. Once the CDF is known, we can compute probabilities using simple rules, such as:

- $P(X < x) = F_X(x)$,
- $P(X > x) = 1 - P(X < x) = 1 - F_X(x)$,
- $P(a < X < b) = F_X(b) - F_X(a)$.

In practice, the choice of which continuous RV to use (e.g., normal, exponential, uniform) depends on which model best fits the real situation we are trying to describe.

6.1 Uniform

The uniform PDF is constant over an interval $[a, b)$. The PDF and its corresponding CDF are

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a \le x < b \\ 0, & \text{elsewhere} \end{cases} \tag{41}$$

$$F_X(x) = \begin{cases} \frac{x-a}{b-a}, & a \le x < b \\ 1, & x \ge b \end{cases} \tag{42}$$

Note 5. Since this is a continuous random variable, the interval over which the PDF is nonzero can be open or closed on either end.

6.2 Exponential

- The probability density function (PDF):
  Let X be an exponential random variable with parameter $\lambda > 0$. We write $X \sim \text{Exp}(\lambda)$.

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \ge 0, \\ 0, & x < 0. \end{cases}$$

- The cumulative distribution function (CDF)

$$F_X(x) = Pr(X \leq x) = \begin{cases} 1 - e^{-\lambda x}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

- Parameters of the Exponential RV:
  For the exponential random variable, the parameter λ is called the ***rate parameter***, and it controls how "fast" events happen. So larger λ means events happen more frequently (shorter waiting times). Smaller λ means events happen less frequently.

- Application:
  Generally the exponential distribution describes waiting time between Poisson occurrences.

Example 6.1. Exponential random variable for inter-arrival times to a server

An e-commerce company operates a customer-support web server that handles incoming chat requests from users. Experience shows that the *time between consecutive chat requests* can be modeled as a continuous random variable X (in minutes) with an exponential distribution. During peak hours, the server receives on average 20 chat requests per hour.

1. (**Model identification**)
   (a) Find the rate parameter λ (in requests per minute).
   (b) Write down the pdf $f_X(x)$ and cdf $F_X(x)$ of the inter-arrival time X.
2. (**Basic probabilities**)
   (a) What is the probability that the next request arrives within 2 minutes?
   (b) What is the probability that you will wait more than 5 minutes for the next request?
3. (**Using the cdf on an interval**) Express $P(2 < X \leq 5)$ in terms of the cdf $F_X(x)$, and then compute its value.
4. (**Memoryless property**) Suppose you have already been waiting 3 minutes and no request has arrived yet.
   (a) Compute $P(X > 5)$.
   (b) Compute the conditional probability $P(X > 5 \mid X > 3)$.
   (c) Compare your answers and explain what they show about the exponential distribution.
5. (**Quantile / design question**) The system administrator wants to set an expectation on the company website: "With 90% probability, your chat request will be picked up within T minutes." Assuming that the waiting time is dominated by the inter-arrival time of new requests, find T such that $P(X \leq T) = 0.9$.

6. (**Designing the rate to meet a quality requirement**) During off-peak hours, the arrival rate may be different and is not known in advance. The company specifies the following quality requirement: "With at least 95% probability, the time between two consecutive chat requests should not exceed 4 minutes." Assuming that off-peak inter-arrival times are still exponential with an (unknown) rate parameter λ_{off}, determine the value of λ_{off} that meets this requirement.

7. (**Connection to Poisson counts**) Let $N(10)$ be the number of requests that arrive in a 10-minute interval. Under the same model:

   (a) What is the distribution of $N(10)$? (Name it and give its parameter.)

   (b) Find $P\big(N(10) = 0\big)$ (no requests in 10 minutes).

   (c) Find $P\big(N(10) \geq 3\big)$.

Note. The same exponential inter-arrival model is commonly used in many other real-world settings, such as arrivals of planes at an airport runway, customers entering a bank or supermarket, phone calls to a call center, or data packets arriving at a network router.

Solution.

We are told that the server receives 20 requests per hour, on average. That is

$$20 \text{ requests/hour} = \frac{20}{60} = \frac{1}{3} \text{ requests/minute.}$$

1. **Model identification.**

   (a) The rate parameter is
   $$\lambda = \frac{1}{3} \quad \text{(request per minute).}$$

   (b) If X is exponentially distributed with rate $\lambda = \frac{1}{3}$, then the pdf is
   $$f_X(x) = \begin{cases} \frac{1}{3}e^{-x/3}, & x \geq 0, \\ 0, & x < 0, \end{cases}$$
   and the cdf is
   $$F_X(x) = P(X \leq x) = \begin{cases} 0, & x < 0, \\ 1 - e^{-x/3}, & x \geq 0. \end{cases}$$

2. **Basic probabilities.**

   (a) Probability that the next request arrives within 2 minutes:
   $$P(X \leq 2) = F_X(2) = 1 - e^{-2/3}.$$

   (b) Probability of waiting more than 5 minutes (i.e., the next request comes after 5 minutes):
   $$P(X > 5) = 1 - P(X \leq 5) = 1 - F_X(5) = 1 - \big(1 - e^{-5/3}\big) = e^{-5/3}.$$

3. **Probability on the interval $2 < X \leq 5$.**

   Using the cdf,

   $$P(2 < X \leq 5) = F_X(5) - F_X(2).$$

   With $F_X(x) = 1 - e^{-x/3}$ for $x \geq 0$,

   $$F_X(5) = 1 - e^{-5/3}, \quad F_X(2) = 1 - e^{-2/3}.$$

   Hence

   $$P(2 < X \leq 5) = \left(1 - e^{-5/3}\right) - \left(1 - e^{-2/3}\right) = e^{-2/3} - e^{-5/3}.$$

4. **Memoryless property.**

   (a) From part (2b),

   $$P(X > 5) = e^{-5/3}.$$

   (b) The conditional probability is

   $$P(X > 5 \mid X > 3) = \frac{P(X > 5 \text{ and } X > 3)}{P(X > 3)} = \frac{P(X > 5)}{P(X > 3)}.$$

   We already know $P(X > 5) = e^{-5/3}$. Also,

   $$P(X > 3) = 1 - F_X(3) = 1 - \left(1 - e^{-3/3}\right) = e^{-1}.$$

   Therefore,

   $$P(X > 5 \mid X > 3) = \frac{e^{-5/3}}{e^{-1}} = e^{-5/3+1} = e^{-2/3}.$$

   (c) Comparison and interpretation:

   Note that

   $$P(X > 5 \mid X > 3) = e^{-2/3} = P(X > 2).$$

   This shows that, given you have already waited 3 minutes, the probability of waiting at least *2 additional minutes* is the same as if you had just started waiting for the first time. This is the *memoryless property* of the exponential distribution:

   $$P(X > s + t \mid X > s) = P(X > t), \quad s, t \geq 0.$$

5. **Quantile / design question.**

   We want T such that

   $$P(X \leq T) = F_X(T) = 0.9.$$

   Using $F_X(T) = 1 - e^{-T/3}$, we solve

   $$1 - e^{-T/3} = 0.9 \implies e^{-T/3} = 0.1 \implies -\frac{T}{3} = \ln(0.1) \implies T = -3\ln(0.1).$$

   Since $\ln(0.1) = -\ln(10)$,

   $$T = 3\ln(10) \approx 6.91 \text{ minutes}.$$

   Thus, with 90% probability, the next request arrives within about 6.9 minutes.

6. **Designing the rate to meet a quality requirement.**

We assume that, during off-peak hours, the inter-arrival time X (in minutes) is exponentially distributed with an (unknown) rate parameter λ_o:

$$X \sim \text{Exp}(\lambda_o), \qquad F_X(x) = P(X \le x) = 1 - e^{-\lambda_o x},\ x \ge 0.$$

The quality requirement says:

$$P(X \le 4) \ge 0.95.$$

Using the cdf,

$$P(X \le 4) = 1 - e^{-4\lambda_o} \ge 0.95.$$

Solve for λ_o:

$$1 - e^{-4\lambda_o} \ge 0.95 \implies e^{-4\lambda_o} \le 0.05.$$

Take natural logarithms:

$$-4\lambda_o \le \ln(0.05) \implies \lambda_o \ge -\frac{1}{4}\ln(0.05).$$

Since $0.05 = \dfrac{1}{20}$, we can write

$$\lambda_o \ge \frac{1}{4}\ln(20) \approx 0.749 \text{ requests per minute.}$$

In terms of requests per hour,

$$\lambda_o \approx 0.749 \times 60 \approx 45 \text{ requests per hour.}$$

Thus, the arrival rate during off-peak hours must satisfy

$$\boxed{\lambda_o \ge \frac{1}{4}\ln(20) \approx 0.75 \text{ per minute (about 45 requests per hour).}}$$

Comment on the original rate and comparison. In the original peak-hours model, the server receives on average 20 requests per hour, which corresponds to

$$\lambda = \frac{20}{60} = \frac{1}{3} \approx 0.333 \text{ requests per minute.}$$

For this original rate, the probability that the time between two consecutive chat requests does not exceed 4 minutes is

$$P(X \le 4) = 1 - e^{-\lambda \cdot 4} = 1 - e^{-\frac{4}{3}} \approx 1 - 0.2636 \approx 0.7364.$$

So, with the original rate $\lambda = \frac{1}{3}$ per minute,

$$P(X \le 4) \approx 0.74,$$

whereas the quality requirement asks for

$$P(X \leq 4) \geq 0.95.$$

This shows that, relative to the desired 95% level, the probability with the original rate is significantly lower and does *not* meet the service-quality target.

By increasing the rate to $\lambda_o \approx 0.75$ per minute (about 45 requests per hour), we obtain

$$P(X \leq 4) = 1 - e^{-4\lambda_o} \approx 1 - e^{-3} \approx 0.95,$$

which satisfies the requirement.

Practical discussion.

Using the original rate $\lambda = \frac{1}{3}$ (corresponding to 20 requests per hour), the probability of waiting less than 4 minutes is

$$P(X \leq 4) = 1 - e^{-4\lambda} \approx 1 - e^{-4/3} \approx 0.736.$$

This value is clearly far below the desired service level of 0.95. In other words, with the original rate, only about 73.6% of inter-arrival times are shorter than 4 minutes, whereas the quality requirement demands at least 95%. To meet this requirement, the system must operate with a higher average rate λ.

In this type of exponential inter-arrival model, a larger rate λ means that events (chat requests, service completions, etc.) occur more frequently on average. From an operational perspective, achieving an *effective* higher rate is often associated with:

- increasing the available **manpower** (for example, assigning more support agents so that multiple chats can be handled simultaneously);
- improving **efficiency** (through better training, standardized procedures, or smarter routing of requests to the most suitable agent);
- enhancing **system capacity** (using faster servers, improved load balancing, or deploying chatbots to automatically answer routine queries);
- reducing overhead and delays via **automation** (automated ticket creation, canned responses, and a more intuitive user interface to reduce back-and-forth).

Although the *arrival rate* of users is primarily driven by customer behavior, in many real systems the "effective" rate at which requests are acknowledged and processed (or the rate of completed services) can be increased by investing in resources and process improvements. The exponential/Poisson model provides a useful bridge between abstract performance requirements such as

$$P(\text{wait} \leq 4 \text{ minutes}) \geq 0.95$$

and concrete design decisions regarding staffing levels, server capacity, and operational policies. In this way, probabilistic modeling helps engineers and managers quantitatively evaluate whether their current configuration can achieve the targeted quality of service and how much additional capacity is needed if it does not.

7. **Connection to Poisson counts.** Let $N(10)$ be the number of requests that arrive in a 10-minute interval.

   For an exponential inter-arrival time with rate $\lambda = \frac{1}{3}$ per minute, the corresponding count of arrivals in time t follows a Poisson distribution with parameter λt.

   (a) Here,
   $$\lambda t = \frac{1}{3} \times 10 = \frac{10}{3}.$$
   So
   $$N(10) \sim \text{Poisson}\left(\frac{10}{3}\right).$$

   (b) Probability of no requests in 10 minutes:
   $$P\big(N(10) = 0\big) = e^{-\lambda t}\frac{(\lambda t)^0}{0!} = e^{-10/3}.$$

   (c) Probability of at least 3 requests:
   $$P\big(N(10) \geq 3\big) = 1 - P\big(N(10) \leq 2\big) = 1 - \big[P(N=0) + P(N=1) + P(N=2)\big].$$
   Let $\mu = \frac{10}{3}$. Then
   $$P(N=0) = e^{-\mu},$$
   $$P(N=1) = e^{-\mu}\frac{\mu^1}{1!} = e^{-\mu}\mu,$$
   $$P(N=2) = e^{-\mu}\frac{\mu^2}{2!} = e^{-\mu}\frac{\mu^2}{2}.$$
   Hence
   $$P\big(N(10) \geq 3\big) = 1 - e^{-\mu}\left(1 + \mu + \frac{\mu^2}{2}\right), \quad \mu = \frac{10}{3}.$$

Example 6.2. Wireless Link Reliability Analysis in a Rayleigh Fading Channel

A wireless engineer is designing a point-to-point communication link for an IoT system. The receiver's **instantaneous signal-to-noise ratio (SNR)**, denoted by X (in linear scale), is random due to multipath fading and is modeled as an **exponential random variable** with probability density function (PDF):
$$f_X(x) = \begin{cases} \frac{1}{\sigma}e^{-x/\sigma}, & x \geq 0, \\ 0, & x < 0, \end{cases}$$
where $\sigma > 0$ is the average SNR.

From field measurements, it is estimated that the **average SNR** is $\sigma = 5$.

The engineer needs to evaluate the system's performance by answering the following:

1. **CDF derivation:** Find the cumulative distribution function (CDF) $F_X(x)$.
2. **Outage probability:** Calculate $P(X < 3)$, the probability that the SNR falls below the required threshold of 3.
3. **Moderate-SNR range:** Determine $P(2 < X < 8)$.
4. **High-quality link probability:** Compute $P(X > 10)$.

Solution

Step 1: CDF derivation

The CDF is defined as:

$$F_X(x) = \int_{-\infty}^{x} f_X(t)\, dt.$$

Since $f_X(t) = 0$ for $t < 0$:

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 1 - e^{-x/\sigma}, & x \geq 0. \end{cases}$$

Substituting $\sigma = 5$:

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 1 - e^{-x/5}, & x \geq 0. \end{cases}$$

Step 2: Outage probability

$$P(X < 3) = F_X(3) = 1 - e^{-3/5}.$$

Numerically:

$$P(X < 3) \approx 1 - e^{-0.6} \approx 1 - 0.5488 \approx 0.4512.$$

Step 3: Probability in range $2 < X < 8$

$$P(2 < X < 8) = F_X(8) - F_X(2).$$

$$= \left[1 - e^{-8/5}\right] - \left[1 - e^{-2/5}\right] = e^{-0.4} - e^{-1.6}.$$

Numerically:

$$\approx 0.6703 - 0.2019 = 0.4684.$$

Step 4: Probability $X > 10$

$$P(X > 10) = 1 - F_X(10) = e^{-10/5} = e^{-2} \approx 0.1353.$$

Final Results

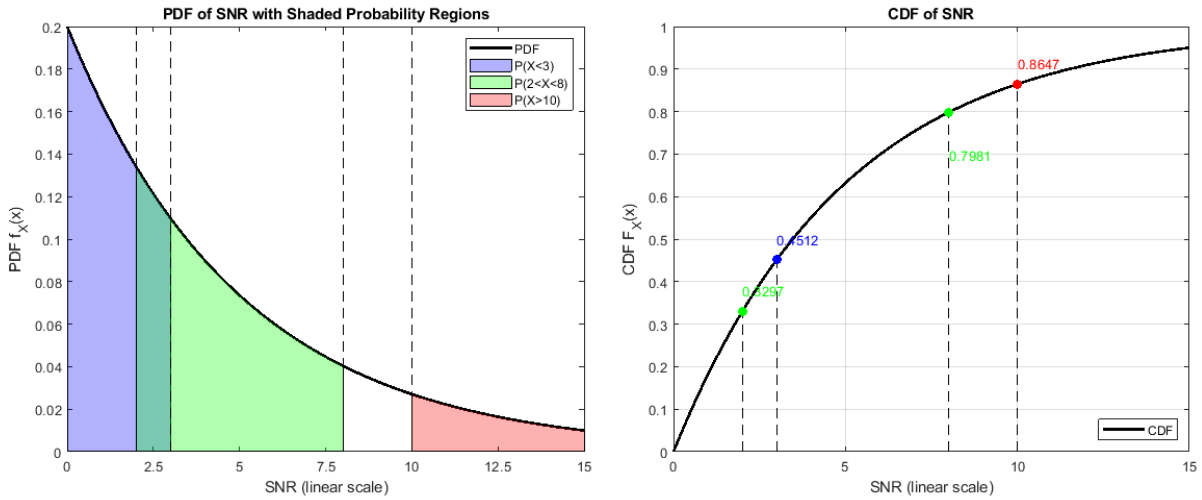


Figure 10: Wireless Link Reliability Analysis (Exponential SNR with $\lambda = 5$)

Event	Probability
$X < 3$ (Outage)	0.4512
$2 < X < 8$	0.4684
$X > 10$	0.1353

The PDF and CDF representation of the above events is depicted in Figure 10.

Engineering Significance

- The outage probability ($P(X < 3)$) indicates that the link will fail to meet the minimum SNR requirement about 45% of the time. This may require improving antenna gain, transmit power, or modulation robustness.
- The moderate-SNR probability ($P(2 < X < 8)$) represents typical operating conditions.
- The high-SNR probability ($P(X > 10)$) shows that excellent link quality occurs around 13% of the time, potentially enabling higher data rates.

6.3 Laplace Random Variable

- A Laplace random variable has a PDF which takes the form of a two-sided exponential.
- PDF

$$f_X(x) = \frac{1}{2b} \exp\left(-\frac{|x|}{b}\right) \tag{43}$$

- CDF

$$F_X(x) = \begin{cases} \frac{1}{2}\exp\left(\frac{x}{b}\right) & ; x < 0 \\ 1 - \frac{1}{2}\exp\left(-\frac{x}{b}\right) & ; x \geq 0 \end{cases} \tag{44}$$

- The width of the PDF is determined by the parameter b, while the peak value of the PDF is $1/2b$.

- Application:
  The Laplace random variable has been used to model the amplitude of a speech (voice) signal.

6.4 Gamma Random Variable

- PDF

$$f_X(x) = \frac{(x/b)^{c-1}\exp(-x/b)}{b\Gamma(c)}u(x) \tag{45}$$

  for any $b,\ c > 0$, and the gamma function Γ is a generalization of the factorial function defined by

$$\Gamma(\alpha) = \int_0^\infty \mathrm{e}^{-t}t^{\alpha-1}\ \mathrm{d}t \tag{46}$$

- CDF

$$F_X(x) = \frac{\gamma(c, x/b)}{\Gamma(c)}u(x), \tag{47}$$

  where $\gamma(\alpha, \beta)$ represents the incomplete gamma function is given by

$$\gamma(\alpha, \beta) = \int_0^\beta \mathrm{e}^{-t}t^{\alpha-1}\ \mathrm{d}t \tag{48}$$

- Application:
  The Gamma random variable (Gamma distribution) appears in many modeling and engineering contexts, including:
  - **Biology and medicine:** Modeling survival times, incubation periods, or time until recovery when the process consists of several stages.
  - **Queueing systems:** The total service time of a customer passing through several exponential service stages (e.g., multiple processing steps or servers).
  - **Reliability and lifetime modeling:** Modeling the lifetime of components or systems that fail after several "damage" events or after accumulating stress over time.

6.5 Erlang Random Variable

- The Erlang random variable is a special case of the gamma random variable.

- PDF

$$f_X(x) = \frac{(x/b)^{n-1}\exp(-x/b)}{b(n-1)!}u(x), \tag{49}$$

  for positive integer n and any $b > 0$.

- CDF

$$F_X(x) = \left[1 - \exp\left(-\frac{x}{b}\right) \sum_{m=0}^{n-1} \frac{(x/b)^m}{m!}\right] u(x) \tag{50}$$

- Application:

Practical Examples of the Erlang Random Variable. The Erlang random variable is a special case of the Gamma distribution with an integer shape parameter $k \in \mathbb{N}$. It is widely used to model the waiting time until the k-th occurrence of an event in a Poisson process. Some practical examples include:

  - **Call centers (telecommunications):** Let calls arrive to a call center according to a Poisson process with rate λ (calls per minute). Then the waiting time until the k-th call arrives follows an Erlang(k, λ) distribution. For example, the time until the 5th call of the day can be modeled as an Erlang random variable.
  - **Network packet transmission through multiple stages:** Consider a data packet that must pass through k independent processing stages (e.g., buffering, encoding, routing) where each stage has an exponential processing time with the same rate λ. The total processing time of the packet is then Erlang(k, λ).
  - **Wireless networks: total delay over multiple hops:** If a packet traverses k independent wireless hops, and the delay on each hop is exponential with the same rate λ, then the total end-to-end delay can be modeled as an Erlang(k, λ) random variable.
  - **Time to complete a multi-step maintenance task:** Suppose an equipment maintenance procedure consists of k independent steps, each with an exponential duration (e.g., inspection, replacement, testing). If the mean duration of each step is $1/\lambda$, the total maintenance time is Erlang(k, λ).

6.6 Chi-Squared (χ^2) Random Variable

- The Chi-Squared random variable is a special case of the gamma random variable.
- PDF

$$f_X(x) = \frac{x^{c-1} \exp(-x/2)}{2^c \Gamma(c)} u(x) \tag{51}$$

  for positive integer or half-integer values of c.

- CDF

$$F_X(x) = \frac{\gamma(c, x/2)}{\Gamma(c)} u(x) \tag{52}$$

- Application:
  The chi-square distribution is used in several statistical tests such as the common chi-square tests (for goodness of fit of an observed distribution to a theoretical one) and Friedman's analysis of variance by ranks.

6.7 Rayleigh Random Variable

- The Rayleigh random variable, like the exponential random variable, has a one-sided PDF.
- PDF

$$f_X(x) = \frac{x}{\sigma^2} \exp\left(-\frac{x^2}{2\sigma^2}\right) u(x) \tag{53}$$

  for $\sigma > 0$.
- CDF

$$F_X(x) = \left(1 - \exp\left(-\frac{x^2}{2\sigma^2}\right)\right) u(x) \tag{54}$$

- The Rayleigh distribution is described by a single parameter, σ^2, which is related to the width of the Rayleigh PDF. The σ^2 parameter is not to be interpreted as the variance of the Rayleigh random variable.
- The Rayleigh distribution arises when studying the magnitude of a complex number whose real and imaginary parts both follow a zero-mean Gaussian distribution.
- Application:
  The Rayleigh distribution arises often in the study of noncoherent communication systems and also in the study of wireless communication channels, where the phenomenon known as fading is often modeled using Rayleigh random variables.

Example 6.3. Rayleigh random variable for vibration amplitude

In a rotating mechanical system, the peak vibration amplitude (in millimeters) measured at a bearing housing is modeled as a Rayleigh random variable. Let A denote the peak vibration amplitude in a given operating cycle. The probability density function (pdf) of A is

$$f_A(a) = \begin{cases} \dfrac{a}{\sigma^2} \exp\left(-\dfrac{a^2}{2\sigma^2}\right), & a \geq 0, \\ 0, & a < 0, \end{cases}$$

where $\sigma > 0$ is a parameter that depends on the machine condition and load.

Assume that for a particular machine, $\sigma = 0.3$ mm.

1. (**Probability of low vibration**) The machine is considered to be in a "very smooth" state if the vibration amplitude does not exceed 0.2 mm. Compute

$$P(A \leq 0.2).$$

2. (**Risk of high vibration**) For amplitudes larger than 0.6 mm, the vibration is considered potentially harmful for long-term operation. Find

$$P(A > 0.6).$$

3. (**Setting a monitoring threshold**)

We are given that the vibration amplitude A has a Rayleigh distribution with parameter σ, and we want to choose a monitoring threshold a_0 such that

$$P(A > a_0) = 0.01.$$

First, recall the cdf of a Rayleigh random variable:

$$F_A(a) = P(A \le a) = \begin{cases} 0, & a < 0, \\ 1 - \exp\left(-\dfrac{a^2}{2\sigma^2}\right), & a \ge 0. \end{cases}$$

Therefore,

$$P(A > a_0) = 1 - P(A \le a_0) = 1 - F_A(a_0) = 1 - \left[1 - \exp\left(-\frac{a_0^2}{2\sigma^2}\right)\right] = \exp\left(-\frac{a_0^2}{2\sigma^2}\right).$$

We want this to equal 0.01, so we solve:

$$\exp\left(-\frac{a_0^2}{2\sigma^2}\right) = 0.01.$$

Take the natural logarithm on both sides:

$$-\frac{a_0^2}{2\sigma^2} = \ln(0.01).$$

Note that $0.01 = \dfrac{1}{100}$, so

$$\ln(0.01) = \ln\left(\frac{1}{100}\right) = -\ln(100).$$

Thus,

$$-\frac{a_0^2}{2\sigma^2} = -\ln(100) \quad \Longrightarrow \quad \frac{a_0^2}{2\sigma^2} = \ln(100).$$

Solving for a_0:

$$a_0^2 = 2\sigma^2 \ln(100) \quad \Longrightarrow \quad a_0 = \sigma\sqrt{2\ln(100)}.$$

This is the general expression for the threshold in terms of σ.

Numerical value for $\sigma = 0.3$ mm.

Let $\sigma = 0.3$ mm. Then

$$a_0 = 0.3\sqrt{2\ln(100)}.$$

Compute step by step:

$$\ln(100) \approx 4.6052, \qquad 2\ln(100) \approx 9.2104,$$

$$\sqrt{2\ln(100)} \approx \sqrt{9.2104} \approx 3.035.$$

Thus,

$$a_0 \approx 0.3 \times 3.035 \approx 0.9105 \text{ mm}.$$

So, choosing a threshold of about

$$\boxed{a_0 \approx 0.91 \text{ mm}}$$

ensures that, under the assumed Rayleigh model with $\sigma = 0.3$ mm, the probability that the vibration amplitude exceeds this threshold is approximately 1%.

6.8 Rician Random Variable

- A Rician random variable is closely related to the Rayleigh random variable (in fact the Rayleigh distribution is a special case of the Rician distribution).
- PDF

$$f_X(x) = \frac{x}{\sigma^2} \exp\left(-\frac{x^2 + a^2}{2\sigma^2}\right) I_0\left(\frac{ax}{\sigma^2}\right) u(x) \tag{55}$$

for and $a,\ \sigma > 0$. In this expression, the function $I_0(x)$ is the modified Bessel function of the first kind of order zero which is defined by

$$I_0(x) = \frac{1}{2\pi} \int_0^{2\pi} \mathrm{e}^{x\cos(\theta)} \mathrm{d}\theta$$

- CDF
  Like the Gaussian random variable, the CDF of a Rician random variable cannot be written in closed form. Similar to the Q -function that is used to describe the Gaussian CDF, there is another function known as Marcum's Q -function which describes the CDF of a Rician random variable. It is defined by

$$Q(\alpha, \beta) = \int_\beta^\infty z \exp\left(-\frac{\left(z^2 + \alpha^2\right)}{2}\right) I_0(\alpha z) \mathrm{d}z \tag{56}$$

  The CDF of the Rician random variable is then given by:

$$F_X(x) = 1 - Q\left(\frac{a}{\sigma}, \frac{x}{\sigma}\right) \tag{57}$$

- The Rician distribution is described by two parameters, a and σ, which are related to the center and width, respectively, of the PDF. As with the Rayleigh random variable, the parameter is not to be interpreted as the variance of the Rician random variable.
- Application:
  The Rician distribution arises in the study of noncoherent communication systems and also in the study of satellite communication channels, where fading is modeled using Rician random variables.

7 Conditional Distribution and Density Functions

In Chapter 2, we defined the notion of conditional probability. In a similar manner, it is quite common to talk about the distribution or density of a random variable conditioned on some event, A.

7.1 Conditional CDF

- Definition of the conditional CDF:

Definition 13 (Conditional CDF)**.** The conditional CDF of a random variable, X, conditioned on the event A having occurred is

$$F_{X|A}(x) = \Pr(X \leq x \mid A) = \frac{\Pr(\{X \leq x\}, A)}{\Pr(A)} \tag{58}$$

Naturally, this definition requires the caveat that **the probability of the event A must not be zero.**

- Properties of conditional CDFs
  The properties of CDFs listed in Equations (22-a) - (22-f) also apply to conditional CDFs, resulting in the following properties of conditional CDFs:

$$F_{X|A}(-\infty) = 0. \tag{59a}$$
$$F_{X|A}(\infty) = 1. \tag{59b}$$
$$0 \leq F_{X|A}(x) \leq 1. \tag{59c}$$
$$\text{if } x_1 < x_2, \text{ then } F_{X|A}(x_1) \leq F_{X|A}(x_2). \tag{59d}$$
$$\text{if } x_1 < x_2, \Pr(x_1 < X \leq x_2 \mid A) = F_{X|A}(x_2) - F_{X|A}(x_1) \tag{59e}$$

- Methods of defining conditioning event A
  In general, there are two methods to define the conditioning event A:
  - Method 1: in terms of the same random variable X (i.e., $A = \{a < X \leq b\}$). This case will be further discussed in this section.
  - Method 2: in terms of another random variable Y (i.e., $A = \{a < Y \leq b\}$). This case will be discussed in next chapters.

- Conditional CDF when $A = \{a < X \leq b\}$
  Suppose that for some arbitrary random variable X, the conditioning event is of the form $A = a < X \leq b$ for some constants $a < b$. Then based on the definition of the conditional CDF (58):

$$F_{X|\{a<X\leq b\}}(x) = \frac{\Pr(X \leq x, a < X \leq b)}{\Pr(a < X \leq b)} \tag{60}$$

  Now, we have the following cases:

– If $x \leq a$, then the cvents $\{X \leq x\}$ and $a < X \leq b$ are mutually exclusive and the conditional CDF is zero.

– For $x > b$ the event $a < X \leq b$ is a subset of $\{X \leq x\}$ and hence $\Pr(X \leq x, a < X \leq b) = \Pr(a < X \leq b)$ so that the conditional CDF is 1.

– When $a < x \leq b$ then $\{X \leq x\} \cap \{a < X \leq b\} = a < X \leq x$ and $\Pr(X \leq x, a < X \leq b) = \Pr(a < X \leq x)$. This can be written in terms of the CDF (unconditional) of X as $\Pr(a < X \leq x) = F_X(x) - F_X(a)$ Likewise, $\Pr(a < X \leq b) = F_X(b) - F_X(a)$.

Putting these results together gives

$$F_{X|\{a \leq X < b\}}(x) = \begin{cases} 0, & x \leq a \\ \frac{F_X(x) - F_X(a)}{F_X(b) - F_X(a)}, & a < x \leq b \\ 1, & x > b \end{cases} \tag{61}$$

Example 7.1. Suppose a random variable X is uniformly distributed over the interval [0,1) so that its CDF is given by

$$F_X(x) = \begin{cases} 0, & x < 0 \\ x, & 0 \leq x \leq 1 \\ 1, & x > 1 \end{cases}$$

Find the conditional CDF of X given that $X < 1/2$.

Solution:

Here, the event $A = \{X < 1/2\}$ is related to a numerical condition on the random variable itself. From the definition of a conditional CDF,

$$F_{X|\{X<1/2\}}(x) = \frac{\Pr(X \leq x, X < 1/2)}{\Pr(X < 1/2)}$$

For $x < 0$, the event $X \leq x$ has probability zero and hence $F_{X|\{X<1/2\}}(x) = 0$ for $x < 0$.

When $0 \leq x \leq 1/2$, the intersection of the events $X \leq x$ and $X < 1/2$ is simply the event $X \leq x$, so that

$$F_{X|\{X<1/2\}}(x) = \frac{\Pr(X \leq x)}{\Pr(X < 1/2)} = \frac{x}{1/2} = 2x \text{ for } 0 \leq x \leq 1/2$$

Finally, for $x > 1/2$, the intersection of the events $X \leq x$ and $X < 1/2$ is simply the event $X < 1/2$ and the conditional CDF reduces to one. Putting this all together, the desired conditional CDF is

$$F_{X|\{X<1/2\}}(x) = \begin{cases} 0, & x < 0 \\ 2x, & 0 \leq x \leq 1/2 \\ 1, & x > 1/2 \end{cases}$$

7.2 Conditional PDF

- Definition:

Example 7.2. The conditional PDF of a random variable X conditioned on some event A is

$$f_{X|A}(x) = \lim_{\varepsilon \to 0} \frac{\Pr(x \leq X < x + \varepsilon \mid A)}{\varepsilon} \tag{62}$$

- Properties of conditional PDF:
  The properties of regular PDFs apply to conditional PDFs as well. In particular,

$$f_{X|A}(x) \geq 0 \tag{63a}$$

$$f_{X|A}(x) = \frac{\mathrm{d}F_{X|A}(x)}{\mathrm{d}x} \tag{63b}$$

$$F_{X|A}(x) = \int_{-\infty}^{x} f_{X|A}(y)\mathrm{d}y \tag{63c}$$

$$\int_{-\infty}^{\infty} f_{X|A}(x)\mathrm{d}x = 1 \tag{63d}$$

$$\int_{a}^{b} f_{X|A}(x)\mathrm{d}x = \Pr(a < X \leq b \mid A) \tag{63e}$$

- Conditional PDF when $A = \{a < X \leq b\}$
  Furthermore, the result in Equation (61) can be extended to the conditional PDF by applying Equation (63 b). This results in the following general formula for the conditional PDF of a random variable, X, when the conditioning event is of the nature $A = \{a \leq X < b\}$:

$$f_{X|\{a \leq X < b\}}(x) = \begin{cases} \frac{f_X(x)}{\Pr(a \leq X < b)}, & a \leq x < b \\ 0, & \text{otherwise} \end{cases} \tag{64}$$

Example 7.3. Let X be a random variable representing the length of time we spend waiting in the grocery store checkout line. Suppose the random variable X has an exponential PDF given by

$$f_X(x) = (1/c)\exp(-x/c)u(x),$$

where $c = 3$ min. What is the PDF for the amount of time we spend waiting in line given that we have already been waiting for 2 min?

Solution:
Here the conditioning event is of the form $X > 2$. We can use the result in Equation (3.41) by taking $a = 2$ and $b = \infty$. The probability of the conditioning event is $\Pr(X > 2) = 1 - F_X(2) = \exp(-2/3)$ Therefore, the conditional PDF is

$$f_{X|\{X>2\}}(x) = \exp(2/3) f_X(x) u(x-2) = \frac{1}{3}\exp\left(-\frac{x-2}{3}\right) u(x-2)$$

Note: It is curious to note that for this example, $f_{X|\{X>2\}}(x) = f_X(x-2)$. That is, given that we have been waiting in line for 2 min, the PDF of the total time we must wait in line is simply shifted by 2 min. This interesting behavior is unique to the exponential distribution and we might not have seen the same result if we had started with a different distribution. For example, try working the same problem starting with a Rayleigh distribution.

7.3 Obtaining the unconditional CDF/PDF from conditional CDFs/PDFs

- Unconditional CDF:
  Consider a set of mutually exclusive and exhaustive conditioning events, $A_1, A_2, \ldots, A_N$. Suppose we had access to the conditional CDFs, $F_{X|A_n}(x)$ $n = 1, 2, \ldots, N$, and wanted to find the unconditional CDF, $F_X(x)$. According to the Theorem of Total Probability

$$F_X(x) = \Pr(X \leq x) = \sum_{n=1}^{N} \Pr(X \leq x \mid A_n) \Pr(A_n) = \sum_{n=1}^{N} F_{X|A_n}(x) \Pr(A_n) \tag{65}$$

  Note 6. The CDF of (unconditional) can be found by forming a weighted sum of the conditional CDFs with the weights determined by the probabilities that each of the conditioning events is true.

- Unconditional PDF:
  By taking derivatives of both sides of equation (65), a similar result is obtained for conditional PDFs

$$f_X(x) = \sum_{n=1}^{N} f_{X|A_n}(x) \Pr(A_n) \tag{66}$$